Molecular Modelling of Lithium Salts in the Binary Mixtures of Ethylene and Ethyl Methyl Carbonates

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SUMMARY

A study of 1M and 3.5M lithium hexafluorophosphate ($LiPF_6$) and lithium bis(trifluoromethanesulfonyl)imidate (LiTFSI) in the binary mixtures of ethyl methyl carbonate (EMC) and ethylene carbonate (EC) with volume ratio of EC:EMC = 0.44 electrolyte systems for Li-ion batteries is done using *ab initio* molecular dynamics (AIMD) and classical molecular dynamics (MD) techniques. To validate theoretical results, this investigation was supported by experimental electrochemical data and small angle X-ray scattering (SAXS) measurements.

A structure factor analysis of systems for both the simulation methods were obtained and compared against SAXS experiment for the purpose of direct validation of theoretical techniques, as well as other structural properties considered in this study. A comparison between LiTFSI and $LiPF_6^-$based electrolytes in the binary mixtures of carbonates using both AIMD and classical MD simulation methods is made. Increasing Li-salt content led to the strengthening of interionic interaction, with the nature of the anion weakly affecting Li-solvent interactions.

The study provides valuable insights into the behavior of LiTFSI and $LiPF_6^-$ based electrolytes in the binary mixtures of carbonates and highlights the importance of understanding the Li^+ solvation behavior and hydrogen bonding interactions in electrolyte solutions to improve the performance of batterie

I. Introduction

A. Lithium-ion Battery

Li-ion batteries (LIBs) have captured the attention of both government and industries over the past decade due to their portable technology, which makes them a promising alternative to gasoline, significantly reducing greenhouse gas emissions. These batteries contain lithium salts, which offer wide operating temperature ranges and high ionic conductivities, making them highly desirable. The basic principle of LIBs is that lithium ions flow back and forth between the electrolytes, aided by the chemical potential difference between the anode's high chemical potential and the cathode's low chemical potential. The discharge of a battery is driven by the flow of Li^+ with the help of the electrolyte (usually an organic solvent) and a separator, as well as electrons form the external circuit. Upon applying a potential difference between the electrolytes, this action is reversed, charging the battery. The capacity of the battery is determined by the reversibility of electron transfer and the requirement for a lower chemical potential to achieve a higher battery voltage. [1, 2]

The practical usage of LIBs is impeded by the growth of Li dendrites, which refers to the formation of dead Li during Li stripping. This phenomenon leads to the consumption of a significant amount of electrolytes for repairing the solid electrolyte, resulting in faster LIB decay and reduced efficiency. To mitigate this issue, an effective alternative is to use of salts containing fluorine atoms in their structure, which enhance cathode performance and alter Li growth patterns. Li salts with perfluorinated anions, such as hexafluorophosphate (PF_6^-), tetrafluoroborate (BF_4^-), trifluoromethanesulfonate (TFO^-), bis(trifluoromethanesulfonyl)imide ($TFSI^-$) *etc.*, are commonly used in LIBs because of their unique balance of good ionic conductivity and ability to passivate aluminum current collectors. Perfluorinated anions aid in the reduction of desolvation energy barriers [1, 3]. Furthermore, halogenation is another well-known method for reducing a substance's flammability, with fluorine being the most compatible halogen for Li-ion batteries due to its powerful free radical scavenging properties [4].

B. About Electrolytes

A major discuss on quality of electrolytes is happening in battery industry. Li salts and solvents media must be of maximum purity. Ethylene carbonate (EC), dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), propylene carbonate (PC) are the most commonly used solvents as an electrolyte. On the other hand, in such media inorganic compounds like $LiPF_6$, $LiBF_4$, LiTFSI are few of the salts providing reasonably high ion conductivity. Generally high dissolution constant Li salts are considered [1]. Solvents used are alkyl carbide because of their anionic stability, low toxicity, a suitable temperature range between the boiling and freezing points and, safety features. Among salts $LiPF_6$ is better as other salts have many problems like lithium perchlorate ($LiClO_4$) is explosive, lithium hexafluoroarsenate ($LiAsF_6$) is poisonous [2]. However, the PF_6^- is unstable thermally and decomposes to LiF and PF_5 that can cause harmful reactions to occur on the surfaces of the electrodes. One of the better alternatives is LiTFSI salt which is known for its safety and stability. But its suspected that LiTFSI is corrosive to aluminum collector [3]. LIBs perform poorly at high temperatures and have a short cycle life because of surface phenomena with both electrodes that enhance impedance while cycling. As a result, in latest years, there has been a significant effort to bring in new solvents, salts, and additives that may result in improved performance of target devices. [2]

Major properties that should be take care of before selecting a solvent is electrochemical stability, operating temperature ranges and safety. To enhance the conductivity of Li salts, it is important to optimize the concentration of salts when using a combination of high polarity, high viscosity solvents, and low polarity, low viscosity solvents. This combination promotes effective conductivity in the system. Temperature ranges mainly effect physical property. Like, high boiling point means high viscosity and polarity and similarly lower freezing point indicates less volatile. So, mixture of alky carbides for solvent helps in achieving nice range of temperatures and can extend the applicability to even lower temperatures. High temperature may affect passivation on electrodes, dangerous interactions or thermal runaway. [2]

To obtain a suitable Li-ion conducting media electrolyte solvent are added. But, pure EC has very high melting point, so DMC or EMC are added to make them suitable. In order to get an electrolyte a

conducting salt is dissolved in the solvent mixture [4]. A crucial element to consider is the solid electrolyte interphase (SEI) layer. The SEI layer characteristics are significantly influenced by the electrolytes compositions, additives, and concentrations. The characteristics of SEI layers are modulated efficiently and effectively by the introduction of certain solvents, Li salts, and additives [5]. Usually, solvents for electrolyte are mixture of two kinds of carbonates, one being cyclic and other being linear. The cyclic ones possessed high salt dissociation and low viscosity for the linear carbonate, which reduces the overall viscosity thereby giving optimum ionic conductivity. EC certainly have high dielectric constant, high thermal stability, and low volatility. It is also reported that EC is stable at even greater potentials around 4.8 V. But, at room temperature it has relatively low ionic conductivity because of its high viscous nature. And for EMC, it certainly have very low viscosity and di electric constant but high volatility [4]. The EC/EMC solvent mixture provides Li-ion a good conducting media considering the mixtures ideal ionic conductivity.

A suitable Li salt is added with the solvent mixture making ion transport possible in electrolyte. There is always an issue of high complexity while adding such salts. If we consider 1M $LiPF_6$ in EC at room temperature, it is liquid in nature, whereas EC purely is solid. They are certain requirements for Li salts in Li ion battery electrolytes that needs to be fulfilled and considered promising electrolytes. It must have the capability of total dissolution and dissociation in the electrolyte for which the salts considered are usually complex anions. The anion's resistance to oxidizing at the cathode electrochemically. They should be chemically inert, mainly very less sensitive to hydrolysis considering if any water impurities are present in the electrolyte. One of the major success reasons for $LiPF_6$ is its capacity to create a durable passive film layer on the surface of the aluminum collector. Electrolyte must be able to withstand the operation temperatures of LIB keeping in mind the internal heat generation during operation. Lastly the electrolytes considered must not have any effect on the environment and health [4] .

C. Background and significance

Many studies have been conducted experimentally and theoretically previously $LiPF_6$-EC-EMC and LiTFSI-EC-EMC of salt electrolyte systems and has brought some promising results in the past. $LiPF_6$ is the essentially sole conducting salt utilized in commercial, cutting-edge LIBs. It has good balance in properties like chemical inertness, ionic conductivity, and passivation of aluminum current collector. This surface phenomenon enhances stability by reducing the oxidation and corrosion. Despite these benefits the salt suffers from safety issues. They have restrictive usage from exceeding 55 ^{0}C temperature as salt begin to degrade. Adding to that $LiPF_6$ salt is highly sensitive to moisture and suffer fire induced toxicity. Tiny amount of moisture may lead to HF formation which is highly toxic compound and also degrades all other components of the cell. But, among all the salts available imide-based salts are one of the promising replacements of $LiPF_6$ commercially. LiTFSI is a best compound from this family of salts. They show better thermal, electrochemical stabilities. Adding to that LiTFSI have outstanding Li solvation and good ionic conductivity. The salt is also less sensitive to moisture which makes it a powerful alternative. Regardless of these all positive features, a major problem is it's in ability to form a passive layer on aluminum current collector thereby not preventing it from being corroded [4].

Tarascon and team report the results of ionic conductivity of various electrolyte composition at different temperatures. It is observed that electrolytes resistance to oxidation was more for $LiPF_6$. At room temperature 1M $LiPF_6$ exhibit an ionic conductivity as high as 11 mS/cm and have wide temperature range from -20 to 100 ^{0}C [6].

To check how the conductivity is varying with concentration Nyman and team conducted an experiment with EC/EMC solvent and $LiPF_6$ salt. They found that at 298 K, the maximum conductivity obtained is 9.5 mS/cm at 1.0 mol/dm^3. A higher conductivity is observed as the concentration of salt increases, particularly at lower concentrations. But, at higher $LiPF_6$ concentrations, due to the high viscosity, conductivity decreases.[7]

In a paper of Botte and team authors conducted a thermal stability experimental study of $LiPF_6$ EC:EMC electrolyte. They considered a temperature range from 50 to 350 ^{0}C. It is observed that as heat is

increased heat consumption of heat production of reaction increases. These high temperatures help to get enough sensitive heat required for the startup of reaction. But the operation temperature should be kept at such extent that it doesn't effect physical property of battery. [8]

Oxidation stability of electrolytes plays an important role in the battery as confirmed by Delp and co-authors who investigated the oxidation stability with different solvents and different salts of an electrolyte. High oxidation stability study helps to understand the electrolytes that can be operated for high voltage cathodes. In general, state of art Li-ion electrolyte can only operate at voltages less than 4.5V. It was anticipated that salts like $LiPF_6$ and LiTFSI would have much higher reduction potentials due to the solvent and anion defluorination processes that lead to LiF production. The reduction and oxidation potentials of both $LiPF_6$ and LiTFSI were similar with around 0.9 and 6.0 E/V respectively. Electrolyte containing $LiPF_6$ in EC/EMC was able to passivate the electrode during reduction scans. [9]

The interparticle interactions between the solvent and ionic species are what cause the electrolyte viscosity. Only with accurate interpretations of different molecules and related parameters is it possible to acquire an in-depth knowledge of viscosity behavior with regard to different solvents, salt concentration, and temperature. Gering predicted electrolyte viscosity $LiPF_6$ EC/EMC system using a molecular-based model. This advanced electrolyte model (AEM), a state-of-the-art modeling tool includes viscosity predictions as part of its comprehensive electrolyte modeling package. At lower temperatures higher concentration salt system have greater viscosities. Similar viscosity trend with respect to salt concentrations are observed at higher temperatures, but overall values of viscosities decrease with the increased temperature This effect based on temperature limits the amount of power that can be produced by LIBs when they are functioning in lower temperatures.[10]

A similar study was conducted by Chidiac and co-workers, where they investigated the effects of salt and solvent on physiochemical properties of electrolyte. They used LiTFSI and LiFSI salts along with mixtures of linear and cyclic carbonates EC/EMC, EC/PC/DMC. They found that LiTFSI is twice as viscous in EM/EMC compared to other solvent. The presence of ions in a solution alters how different species interact with each other, which can lead to changes in the viscosity of the solution. These changes

depend on the type of solvent being used and whether or not there is salt present. Furthermore, the calculation of conductivity and viscosity using BioLogic© multichannel conductometer and Anton Parr digital vibrating tube densitometer, respectively, in LIBs was linked to electrochemical performance, allowing to understand how the ions and solvents in those electrolytes were organized. Ionic conductivity for LiTFSI in EC/EMC system increases as temperature is increased.[11]

A molecular dynamics (MD) simulation using polarizable force field is done for EC with LiTFSI salt by varying salt concentrations by Borodin and team. A major take away from this paper is that the Li is solvated by the carbonyl atoms from EC and as the aggregate size is increased the ions probability of participating in ion aggregates are decreased exponentially.[12]

In a paper of Logan[13] on study of LIB physical chemical properties like viscosity and electrolytic conductivity of $LiPF_6$ in EC/EMC system as a function of temperature. Its observe that these physical and chemical properties can be effected on addition of esters. Furthermore, it was shown that the viscosity of the electrolyte (over concentration from 0 to 2 mol/kg) increases and with decreasing of temperature. On the other hand, conductivity increases when concentration goes from 0 to 1 mol/kg and decreases from 1 mol/kg to 2 mol/kg, while monotonically increases with temperature. A similar kind of study is done by Chidiac[11] in which a similar observation is made with LiTFSI system.

Chen[14] using density function theory (DFT) investigated the interaction contribution solvent-solvent, cation-solvent, cation-anion. He considered EC solvent together with $LiPF_6$ and LiTFSI. They observed that Li^+-EC binding energy is least among others in the solution, similarly they got binding energy of Li^+-PF_6^- in EC is least means having the highest dielectric constant. They also got that binding energy of Li^+-TFSI was greater than Li^+-PF_6 indicating worst and best dissolution of salt to some extent[14].

Nevertheless, despite sufficient lack of knowledge on interionic, intermolecular, and ion-molecular interactions on atomistic level for, we still have many investigation opportunities. Although extensive information on microscopic structural details is difficult to get through experimental approaches, it has been proven that *ab initio* molecular dynamics (AIMD) and classical MD simulations techniques can be a valuable alternative on the investigation of electrolyte systems.

Thus, the main goal of this study to analyze and interpret electrochemical experimental data provided and to investigate the structural properties using computational approach. Two electrolytes have been used at two different concentrations: 1M and 3.5M $LiPF_6$ in EC/EMC, and LiTFSI in EC/EMC. Li-ion solvation environment and hydrogen bonding interactions are studied. Structure factor analysis is used to check the match of theoretical data against the SAXS experimental data.

II. Atomic-scale computation

A. Classical Molecular Dynamics Simulation

Classical MD simulation is a computational method for studying the time evolution of atoms and molecules in a system. It is an extremely useful tool for analyzing the structure, dynamics, and thermodynamic properties of materials ranging from small molecules to large biomolecules and even bulk materials. The method is widely used in a broad spectrum of scientific disciplines, including chemistry, physics, materials science, and biophysics. [2]

Classical MD simulations are based on molecular mechanics principles that also define the motion of atoms and molecules as they interact with one another via a set of interatomic potentials. The essential thought behind a classical MD simulation is clear, given the positions of the entirety of the atom; one can compute the force applied by all the atoms on each individual atom. To predict spatial position as function of time for each atom, Newton's law of motion is used. Specifically, one-step through time, over and over ascertaining the powers on every atom and afterward utilizing those forces to update the velocity and position of each individual atom. The subsequent trajectory is, fundamentally, a three-dimensional picture that depicts the atomic level setup of the framework at each point during the simulated time span.[15]

One of the most crucial methods for investigating the foundations of battery electrolytes, including their bulk/interfacial structures, physiochemical properties, and interfacial interactions, is classical MD simulations. Using statistical analysis, classical MD simulations can link the macroscopic and microscopic characteristics of electrolytes. Since 2010, classical MD simulations have evolved into an extremely significant and nearly essential theoretical tool for the research of battery electrolytes. [16]

The fundamental principle of molecular dynamics is the use of classical mechanics to numerically combine the equations of motion with each particle in a system. Newton's laws of motion, which explain how a particles position, velocity, as well as acceleration alter over time, are utilized to derive the equations of motion. The following are the equations of motion for a particle of mass *m*:

$$F = ma \tag{1}$$

where *F* is the force acting on the particle, and a is the acceleration of the particle. The force acting on the particle can be calculated as negative gradient of a potential energy function, *U*, which describes the interaction between the particles in the system:

$$F = -\nabla U \tag{2}$$

The interaction potential is divided into non bonded and bonded interaction, and initial three terms represent the bonded interaction, considering angle bending, dihedral torsional and bond stretching interactions whereas, last two in the equation are non-bonded interactions including van der Waals interaction (using Lennard-Jones) and electrostatic interactions.[17]

$$U = U_{bond} + U_{angle} + U_{dihedral} + U_{vdw} + U_{coloumb} \tag{3}$$

Microcanonical (constant *nVE*), isothermal-isobaric (constant *npT*), canonical (constant *nVT*), and grand canonical (constant *VT*) ensembles are the four varieties of ensembles that can be classified into the same macroscopic state. *n, V, E, T* and *p*, denote the number of particles, volume, total energy, temperature and pressure. The simulation system needs to include enough particles and run long enough to guarantee an accurate forecast of macroscopic attributes. Yet, the computer power unavoidably places a limit on the system size. Even though more than 30 different types of force fields have been established over the past few decades, generating realistic force fields is an old though still relevant topic in contemporary MD studies. For reasonable MD simulation results, selecting relevant models and suitable force fields is equally important.[16]

B. *Ab Initio* Molecular Dynamics Simulation

AIMD is a computational chemistry method for studying the behavior of atoms and molecules in a dynamic environment. It is based on *ab initio* simulation scheme and developed during 1960's.[18] The potential energy of the system is computed from first principles in this method, without the use of any experimental parameters. The term "first-principles" methods refer to a set of well-developed approximations to the correct quantum-mechanical characterization of a material in the framework of

electronic structure calculations. The motion of the atoms is then simulated by using this potential energy to solve the classical equations of motion.[19]

The majority of conventional force fields have some limitations. The flexibility of the interaction characteristics to various chemical systems is a significant challenge that arises [20]. It is necessary to take a more comprehensive strategy in which the electrical degrees of freedom are expressly taken into consideration. One of the most significant advancements in the field among the suggested solutions to these restrictions has been the integration of MD with first-principle calculations within the AIMD [21].

The Born-Oppenheimer approximation, which separates electron and nuclei motion, is the main equation used in AIMD. According to this approximation, the electronic motion is much faster than the nuclear motion and can thus be treated separately. The electronic wave function is calculated by solving the Schrödinger equation given by:[22] [17]

$$H\Psi = E\Psi \tag{4}$$

where H is the Hamiltonian operator, Ψ is the electronic wave function, E is the energy of the system, and the Hamiltonian is given by: [22]

$$H = -\sum\left(1/2*m_i\right)\nabla i^2 - \sum\left(Ze^2/r_i\right) + \sum\sum\left(e^2/|r_i - r_j|\right) \tag{5}$$

where the first term represents the electrons kinetic energy, the second term signifies the attraction in between electrons as well as the nuclei, and the third term signifies the electrons repulsion. The Schrödinger equation is generally solved numerically using methods including the finite difference or finite element methods. [19]

DFT is the most widely utilized quantum mechanical technique in this area. Any decently sized molecule's energy and wave function must be determined, which is a difficult operation. The Born-Oppenheimer approximation (BOMD) reduces this. The electronic wave function's convergence at each MD phase is the computational constraint in BOMD. The ground-breaking method developed by Car and Perriello in 1985 offered a solution to this issue that was both elegant and precise, and is today known as CPMD [23] . The aforementioned formalisms of BOMD and CPMD have been expanded to include AIMD for the simulation of quantum nuclei.

Both classical and *ab initio* MD simulation have its own pros and cons. The AIMD is extremely time consuming but does not require any empirical knowledge. Whereas the classical molecular dynamics are very fast but depends on potential feasibility. Even though numerical methods and computer system performance are continually improving, ab initio calculations continue to stay computationally expensive. In contrast, classical MD is much quicker and can consider large numbers of atoms. This is mainly owing to the fact that empirical interatomic potentials, in which electronic effects are implicitly integrated. Nevertheless, developing potential parameters that adequately reflect the physical properties of the material under analysis can be difficult. Because AIMD uses no a priori potential, potential sources of uncertainty caused by potential selection and variable fitting can be overlooked.[24]

C. Analysis of MD/AIMD Trajectories

1. Radial Distribution Function

The radial distribution function (RDF) offers information about the local density in proximity to a given point as a function of distance. It is a mathematical function that describes the spatial distribution of particles within a system. The RDF determines the probability of finding a particle at a specific distance from a reference particle, typically measured by the assumed particle density in the system. Experimentally, the same information can be obtained using the X-ray diffraction technique. Then, the Fourier function is used for calculating the structure factors. We can evaluate RDF typically written as $g(r)$ as[25]

$$g(r) = (1/n) * (dN(r) / 4pr^2 dr) \quad (6)$$

where n is the particle density in the system, N(r) is the total number of particles located within a shell of radius r, thickness d_r around in the reference particle, and the factor of $1/4\pi r^2$ is the spherical shell's surface area. The above equation is a generalized equation. [25]

$$g_{ab}(r) = \frac{V}{N_a \cdot N_b} \sum_{i=1}^{N_a} \sum_{j=i+1}^{N_b} \left\langle \delta\left(r - \left|\vec{r}_i(t) - \vec{r}_j(t)\right|\right)\right\rangle_t \quad (7)$$

where r_i and r_j indicate position vector of ith and j^{th} particle, δ takes the values 0 or 1 depending on the bin width.[26]

RDF values greater than one indicate that the probability of finding an identified particle in this distance from the reference particle is higher than it should be on average. RDFs are important because they can be estimated not just from simulations but also based on X-ray and neutron diffraction experiments for certain types of samples. For RDFs the observed and the reference particle can be chosen from different molecules and the plot is of intermolecular nature and also, we can consider the reference and the observed particle from same molecule to get intramolecular information. [26] RDFs are important as they are the main connection between macroscopic thermodynamic properties and intermolecular interactions between mixture and the particle. [26, 27]

2. Coordination Number

Coordination number is number of atoms, ions within nearest-neighbor distance. It can be calculated by using below formula

$$CN = 4\Pi\rho\int_{0}^{R} g(r)r^2 dr \qquad (8)$$

where r is the minimum position right after initial vertex of RDF i.e., g(r), ρ is the density and R is the maximum radius of the particle.[25]

A central atom coordination number can influence its physical and chemical properties, as well as the characteristics of the system as a whole. A variety of methods, including X-ray crystallography, chemical analysis and spectroscopy can be used to determine the coordination number. In X-ray crystallography, for example, the positions of atoms in a crystal can be determined by analyzing the scattering of X-rays transmitted through to the crystal. The coordination number is then calculated by counting the number of atoms surrounding the main atom. It helps in determining the structure and also its physical and chemical properties like thermal conductivity, magnetic behavior. Also, Stability constant is one of the measures of coordination number which is measure of bond strength. [27, 28]

3. Structure Factor

Structure factor is a mathematical measurement used to identify the crystalline material structure. It is characterized by the ratio of actually observed X-ray or neutron diffraction frequency from a sample to expected intensity from crystal. The structure factor is an index of the configuration of atoms inside the crystal, and it can discover important info regarding the positions of the atoms, crystal lattice, and the symmetry of the crystal. It can be used to detect and evaluate errors and imperfections in crystals that really can affect the characteristics and performance of the material. [29, 30]

Bragg's law, which refers the angles where the X-rays or neutrons are diffracted by a crystal to the distance between the atoms in the crystal lattice, is commonly used to determine the structure factor and for each set of lattice planes can be determined by analyzing the intensity of the diffracted beams. One can determine the distances and alignment of atoms in a crystalline structure by monitoring the diffraction pattern of a specimen and estimating its structure factor. This data can be used to determine the material crystal structure and symmetry, which could provide important insights into its chemical and physical characteristics. We can use this data for optimizing the property of material. [29, 31]

The structure factor is a function related to RDF by Fourier transform.

$$S(\mathbf{q}) = 1 + \rho \int e^{-i\mathbf{q}r}(g(r) - 1)\mathrm{d}r \quad (9)$$

where S(**q**) is scattering intensity. In our study we used X-ray scattering cross section. To compute in a simulation trajectory, all atom pairs involve hereby making it to compute for all pairwise RDFs of system and finally summed up. Simulation tools like TRAVIS is used for measuring which helps in giving the outputs directly. [25, 32]

III. Computational details

In order to conduct AIMD simulation each of the system corresponding to 1M and 3.5M Li-salt electrolyte consisting PF_6^- and $TFSI^-$ anions in the binary mixture of EC/EMC with volume ration of 11:25, was preliminary pre-equilibrated using classical MD simulation according to methodology proposed by Crabb *et al*[33]. For this purpose a supercell of 17×17×17 Å was generated using Packmol code [34] and comprising 2 and 7 structure units, so that the number of atoms does not exceed 300 atoms (see Table 1 for details). Herein and below, all MD simulations were carried out using the GROMACS software package, version 2016.3 [35] . All the systems were equilibrated in isothermal-isobaric (*npT*) ensemble for 10 ns, followed by canonical (*nVT*) ensemble simulation for additional 2 ns. To achieve thermodynamic conditions of 298.15 K and 1 bar, the leap-frog algorithm was used to integrate the motion equations, with a time-step of 0.5 fs. and were maintained using velocity-rescaling thermostat [36] and Parrinello-Rahman barostat [36] (see details below). After the systems achieve equilibrium of the potential energy within the classical MD framework, the final configurations were used as the initial structures for the subsequent AIMD simulation.

The AIMD simulations have been performed in the context of the density-functional theory, or DFT, with the projector augmented wave technique using the VASP, "Vienna Ab initio Simulation Program," code [37-39]. The Perdew-Burke-Ernzerhof, PBE, functional [40] was used in the simulations, along with zero damping DFT-D3 correction method, [41] Γ-point Brillouin zone sampling, the plane wave cutoff of 400 eV energy, and convergence threshold of 105 eV energy, and a 0.5 fs time step. All equilibration was done in the nVT ensemble with a Nosé-Hoover thermostat set to 834 cm^{-1} and a duration of 40 fs, however all production runs were done in the microcanonical ensemble (nVE). Each production run lasted minimum 10 ps, with the first pico-second deleted from the final data to allow for additional equilibration as the nVT and nVE ensembles transitioned.

TABLE I

SYSTEM COMPOSITION USED THROUGHOUT THIS WORK, AND EQUILIBRIUM SUPERCELL SIZE AND DENSITIES, *D*, OBTAINED BY MEANS OF AIMD AND CLASSICAL MD SIMULATION AGAINST EXPERIMENTAL MEASUREMENTS [37-40] AT 298.15K.

System	# Li-salt	# EC	# EMC	Supercell / Å	d_{theor} / g cm^{-3}	d_{exp} / g cm^{-3}
			AIMD			
1M $LiPF_6$/EC/EMC	2	9	14	16.1864×16.1864×16.1864	0.8631	1.0620
1M LiTFSI/EC/EMC	2	9	14	16.7385×16.7385×16.7385	0.9545	1.1750
3.5M $LiPF_6$/EC/EMC	7	9	14	17.6542×17.6542×17.6542	1.1199	1.1790
3.5M LiTFSI/EC/EMC	7	9	14	19.1962×19.1962×19.1962	1.4398	1.1900
			MD			
EC/EMC		554	1080	61.6748×61.6748×61.6748	1.1349	1.0550
1M $LiPF_6$/EC/EMC	84	554	1080	62.4957×62.4957×62.4957	1.1878	1.0620
1M LiTFSI/EC/EMC	84	554	1080	63.1521×63.1521×63.1521	1.2144	1.1750
3.5M $LiPF_6$/EC/EMC	294	554	1080	64.0958×64.0958×64.0958	1.2935	1.1790
3.5M LiTFSI/EC/EMC	294	554	1080	66.5859×66.5859×66.5859	1.3753	1.1900

In terms of classical MD simulation, each system was generated using the same tools as for AIMD. For 1M and 3.5M Li-salt electrolyte, the system with the relatively large ensemble of particles comprising 1718 and 1928 structural units, respectively, was generated and placed into the orthorhombic box of 86×86×86 Å with three-dimensional periodic boundary conditions. The selection of such system size was dictated by the attempts to reproduce the entire statistical local structure pattern for all the possible interionic, ion-molecular, and intermolecular interactions within the structure factor analysis and SAXS experiment, respectively. In addition, a neat binary mixture of EC and EMC with the system size of 1634 was generated as well for the purpose of referencing (Table 1). Nevertheless, a separate run with a double-edged box with approximately 10 times larger number of structure units, was conducted for each of the considered systems to account the lower q values in $S(q)$.

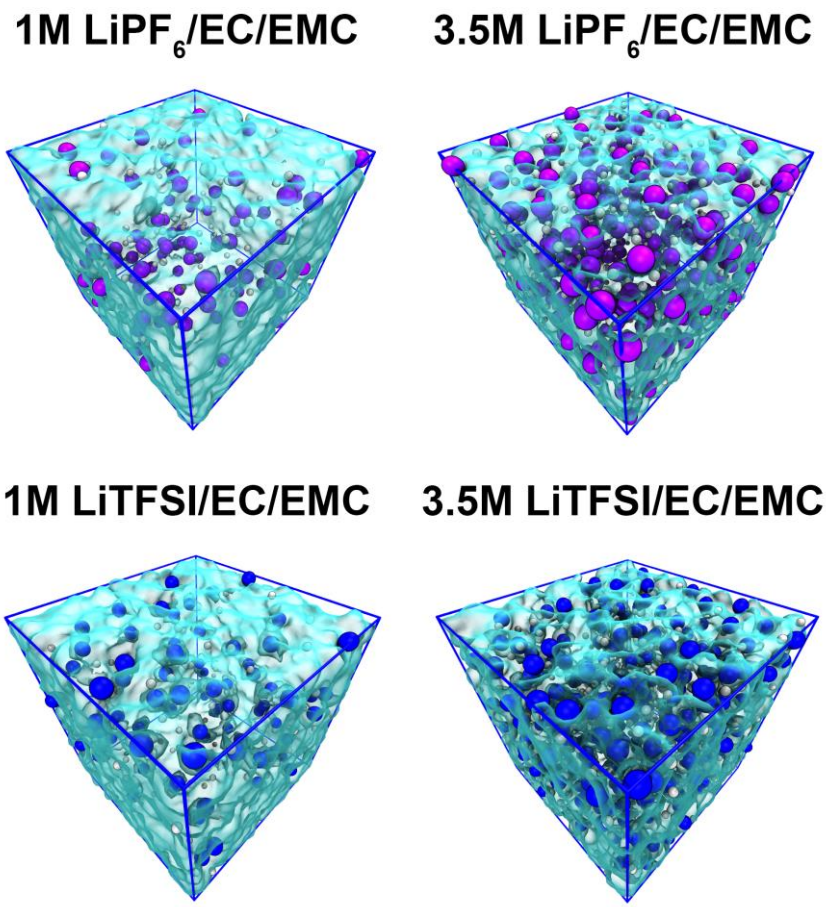


Figure 1. Illustration of the snapshots obtained after the equilibration of the $LiPF_6$/EC/EMC and LiTFSI/EC/EMC supercells by means of classical MD simulation at 298.15K for 3.5 M (*right-hand*) and 1 M (*left-hand*) Li-salt composing PF_6^- (*top*) and $TFSI^-$ (*bottom*) in the binary mixture of EC and EMC with a volume ratio of 11:25.

The potential model variables were assessed in terms of their ability to regenerate the experimental density at ambient temperature. Each system was, firstly, minimized using steepest descent algorithm [41] with the default convergence criterion, following equilibration procedure in *npT* ensemble for 20 ns and *nVT* ensemble for additional 10 ns (Figure 1). To simulate a larger system the simulation time was doubled to achieve equilibrium. After reaching the equilibrium of the potential energy, a production calculation of 20 ns in *nVT* ensemble was additionally conducted to collect the trajectory.

During the equilibration process, the temperature was maintained at a constant level by applying velocity-rescaling thermostat [36]. The pressure was fixed at 1 bar with the Parrinello-Rahman barostat [36], with coupling constants 0.1 ps and 0.5 ps, respectively. During the production stage, a Nosé-Hoover

was employed [47]. The equations of motion were integrated using the leap-frog algorithm with a time-step of 0.5 fs. Long-range electrostatic interactions within the 1.4 nm cutoff range were handled using the computationally efficient Particle-Mesh-Ewald algorithm [48], which also used a similar cutoff distance for the real-space component. The 12-6 Lennard-Jones interactions were treated using the conventional shifted force technique, with a switch region between 1.2 and 1.3 nm. Cross-interactions between different types of atoms were derived with the help of standard Lorentz-Berthelot combination rule. [42].

All the analysis discussed in this work was conducted using TRAVIS, "Trajectory Analyzer and Visualizer" code, version July 29, 2022, [26, 32] and VMD, "Visual Molecular Dynamics", package used for all the structural visualization. [43]

IV. Force Field

Quantum chemistry allows for the first-principles prediction of all the specifics of molecule interactions and structures. However, due to the prohibitively expensive computational requirements, many of the issues we are concerned in are unable to be solved using quantum-chemical calculations. Hence, the study of structure of molecules and dynamics requires the creation of simplifications. Using force field only nuclei motion is calculated, and electronic degree of freedom are not considered. The equations selected to represent the potential energy as well as the accompanying parameters make up a force field. The fact that molecular mechanics force fields are subjective must be remembered. In general, combining energy functions and characteristics from various force fields to describe a specific molecule is incorrect [44].

AMBER ("Assisted Model Building with Energy Refinement") [45], CHARMM ("Chemistry at Harvard Macromolecular Mechanics") [46] , and OPLS ("Optimized Potentials for Liquid Simulations") [47] were among the extensively utilized force fields databases in the early 1980s. OPLS, in contrast to AMBER and CHARMM, set out from the start to replicate properties of a liquid form, such as density and heat of vaporization. This was accomplished by determining nonbonded interaction by modeling simple fluid thermodynamic properties as well as empirical parameters measured from the experiment and adopted to gain transferability between different derivatives of the similar chemical nature. Similar to the other two force fields, OPLS began with a united atom, -UA, model, where the certain groups expressed as a single bead encompassing the individual atomic features in one term, e.g., CH, CH_2, CH_3, CF_2 etc., before switching to an all-atom, -AA, model later on.[48, 49]

As previously stated, the formulation for a molecular potential energy of the system that is most usually used for simple organic compounds is the following: [44]

$$U = \sum_{\text{bonds}} \frac{k_r}{2}(r_{ij} - r_0)^2 + \sum_{\text{angle}} \frac{k_\theta}{2}(\theta_{ijk} - \theta_0)^2 +$$
$$+ \sum_{\text{dihedral}} C_{n=0-5}(\cos(\phi_{ijkl} - 180))^n + \sum_{\text{LJ}} 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] +$$
$$+ \sum_{\text{Coulomb}} \frac{1}{4\pi\varepsilon_0}\frac{q_i q_j}{\varepsilon_r r_{ij}} \qquad (10)$$

where r is the actual bond length. A harmonic potential is typically used to illustrate both, the bond stretching and angle bending terms. For bond, force constant is denoted by k_r, is usually very high, suggesting that it requires a lot of energy to appreciably compress or stress a chemical bond (see Table 2 and 3 for details). The equilibrium bond length, or r_0, is the bond length at a potential energy function with all other terms set to zero. The potential energy shift caused by the deformation of angles is represented by the second term with the respective equilibrium angle Θ_0 and force constant k_θ expressed in a harmonic potential. As a function of the rotations about every dihedral angle, the third term (typically called to as the torsional term) indicates the attainable energy of the molecules. In this case, n corresponds to the number of minima within energy function (also known as multiplicity), and phase factor ϕ_0 determines the position of the minima. The so-called nonbonded or interatomic terms i.e., fourth and fifth terms in Equation 10, which are calculated for pairings of atoms separated by three covalent bonds (e.g., 1-4 interactions) and between atoms in distinct molecules. The van der Waals component of the potential is known as Lennard-Jones 12-6 potential contains repulsive and attractive term [44]. Because various atoms have varying abilities to attract electrons, the electric charge in molecules is not spread uniformly. The total electrostatic potential created by charges applied to atomic nuclei, or partial atomic charges, is one way to visualize the electrostatic potential of a molecule. If one wants to apply Coulomb forces in molecular mechanics simulations, the advantage of atom-centered charges is that they are simple to calculate and operate directly on the nuclei. In the equation q_i and q_j represent the charge on the atoms.

Therefore, an OPLS-AA-based potential parameters were selected to describe $LiPF_6$ and LiTFSI salts, as well as EC and EMC molecular liquids. The Coulomb potentials were revised to reproduce better the electrostatic interactions between the electrolyte constituents, which are the key component in such systems. Thus, second order Møller-Plesset perturbation theory, MP2, functional coupled with consistent basis sets of triple-ζ valence with polarization quality, aug-cc-pVTZ, basis set developed by Dunning *et al*. [50] was used to optimize in gas phase each of the individual chemical structure, followed by electrostatic potential calculations using the same level of theory. The latter was achieved using "Charges from Electrostatic Potentials using a Grid based method", CHELPG, scheme by Breneman and Wiberg [51] as

available in Gaussian code, version 16 [52]. In this method, atomic charges are attached to replicate the molecular electrostatic potential (MEP) at various locations around the molecule. As a first step in the fitting process, the MEP is estimated at a set of grid points that are regularly distributed in a cube and spaced 3.0 pm apart. The cube dimensions are designed by keeping molecule is in the middle of the cube, leaving a headspace of 28.0 pm between the box end and the molecule in all three dimensions. Any places within the molecule van der Waals radius are excluded from the fitting operation. After evaluating the MEP at all acceptable grid points, atomic charges which it best duplicates the MEP are generated. The sole extra constraint in the fitting technique is that the total charge of the system equals the sum of all atomic charges. All the potential parameters including ones derived using quantum-chemical calculations are listed in Table 2 ($LiPF_6$ and LiTFSI) and Table 3 (EC and EMC).

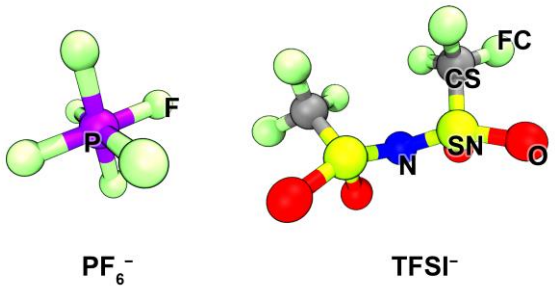


Figure 2. Molecular structure of the perfluorinated anions, hexafluorophosphate PF_6^- and bis(trifluoromethanesulfonyl)imidate $TFSI^-$, composing Li-salts in electrolyte solutions studied in this work and the atomic type labeling used within potential model parameters. Color coding for representation: purple ─ P, lime ─ F, grey ─ C, yellow ─ S, red ─ O, blue ─ N.

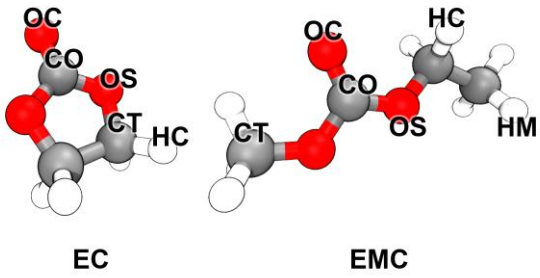


Figure 3. Molecular structure of the molecular solvents, ethylene carbonate EC and ethyl methyl carbonate EMC, used in electrolyte solutions studied in this work and the atomic type labeling used within potential model parameters. Color coding for representation: grey ─ C, red ─ O, white ─ H.

TABLE II.

POTENTIAL MODEL PARAMETERS FOR $LiPF_6$ AND LITFSI USED IN THE FRAMEWORK OF CLASSICAL MD SIMULATION IN THIS WORK (SEE DESCRIPTION FOR THE EQUATION 10).

Atom type	q / e	σ / Å	ε / kJ mol^{-1}
Li	1.0000	1.5100	0.6906
Atom type	q / e	σ / Å	ε / kJ mol^{-1}
N	−0.2800	3.2500	0.7113
SN	0.3100	3.5500	1.0460
O	−0.3200	2.9600	0.8786
FC	−0.1800	2.9500	0.2252
CS	0.5100	3.5000	0.2761

r_0 / Å	k_r / kJ mol^{-1} Å^{-2}
1.5700	31370000.0000
1.4370	53310000.0000
1.5750	18790000.0000
1.8180	19700000.0000
1.3230	36970000.0000

θ_0 / deg	k_θ / kJ mol^{-1} rad^{-2}
113.6000	789.0000
103.0000	902.0000
125.6000	671.0000
104.1000	1077.0000
118.5000	969.0000
100.2000	816.0000
102.6000	870.0000
111.7000	694.0000
107.1000	781.0000

C_0	C_1	C_2	C_3	C_4	C_5 / kJ mol^{-1}
0.0000	0.0000	−0.0150	0.0000	0.0000	0.0000
13.4514	−12.3373	−8.4874	−2.8654	0.0000	0.0000
0.0000	0.0000	1.4510	0.0000	0.0000	0.0000
0.0000	0.0000	1.3520	0.0000	0.0000	0.0000
32.7730	−10.4200	−3.1900	0.0000	0.0000	0.0000

Atom type	q / e	σ / Å	ε / kJ mol^{-1}
P	1.7000	3.7400	0.8368
F	−0.4500	3.1180	0.2552

r_0 / Å	k_r / kJ mol^{-1} Å^{-2}
1.6060	31000000.0000

θ_0 / deg	k_θ / kJ mol^{-1} rad^{-2}
90.0000	1165.0000
180.0000	291.0000

TABLE III.

POTENTIAL MODEL PARAMETERS FOR EC AND EMC USED IN THE FRAMEWORK OF CLASSICAL MD SIMULATION IN THIS WORK (SEE DESCRIPTION FOR THE EQUATION 10).

Atom type	OPLS nomenclature	q / e		σ / Å	ε / kJ mol^{-1}
		EC	EMC		
OS	opls_773	−0.4400	−0.5300	3.0000	0.7113
CT	opls_774	0.2200	0.5200	3.5000	0.2761
HC	opls_777	0.0200	0.0100	2.4200	0.0628
CO	opls_772	0.9900	1.1100	3.7500	0.4393
OC	opls_771	−0.6300	−0.6700	2.9600	0.8786
HM	opls_140		0.0700	2.5000	0.1255

Bond type	r / Å	k_b / kJ mol^{-1} Å^{-2}
CO-OC	1.2290	47697600.0000
CO-OS	1.3270	17907520.0000
OS-CT	1.4100	26777600.0000
CT-HC	1.0900	28451200.0000
CT-CT	1.5290	22426240.0000
CT-HM	1.0900	28451200.0000

Angle type	θ / deg	k_θ / kJ mol^{-1} rad^{-2}
OC-CO-OS	123.4000	694.5440
CO-OS-CT	116.9000	694.5440
OS-CT-HC	109.5000	292.8800
OS-CT-CT	109.5000	418.4000
OS-CO-OS	118.1800	584.9230
HC-CT-HC	107.8000	276.1440
HC-CT-CT	110.7000	313.8000
CT-CT-HM	110.7000	313.8000
HM-CT-HM	107.8000	276.1440

Dihedral type	C_0	C_1	C_2	C_3	C_4	C_5 / kJ mol^{-1}
OC-CO-OS-CT	20.9200	0.0000	−20.9200	0.0000	0.0000	0.0000
CO-OS-CT-HC	0.4142	1.2427	0.0000	−1.6569	0.0000	0.0000
CO-OS-CT-CT	−2.1966	5.2007	0.5272	−3.5313	0.0000	0.0000
HC-CT-CT-HC	0.6276	1.8828	0.0000	−2.5104	0.0000	0.0000
OS-CT-CT-OS	1.1506	1.1506	0.0000	0.0000	0.0000	0.0000
OS-CT-CT-HC	0.9790	2.9372	0.0000	−3.9162	0.0000	0.0000
OS-CT-CT-HM	0.9790	2.9372	0.0000	−3.9162	0.0000	0.0000
CT-OS-CO-OS	31.2060	−9.7680	−21.4390	0.0000	0.0000	0.0000
HM-CT-CT-HC	0.6276	1.8828	0.0000	−2.5104	0.0000	0.0000

V. Results & Discussions

A. AIMD Approach

1. Structure Factor analysis

Structure factor is a key parameter in comparison theoretical and experimental analyses of the structure of material. Here the structure factor analysis is obtained from the equilibrated unit cell using AIMD simulation. The results are then compared with the available SAXS data curves in the Figure 4. Comparing the theoretical and experimental structure results can provide important insights into the structure of the material being studied. From the graphs we can say that structure factor obtained from AIMD is not matching with experimental structure factor. Checking the corresponding shift of shoulder with increase in concentration for both the salt systems, its observed that increase in salt concentration promotes a shift of the peak Δq by 0.12 and 0.22 $Å^{-1}$ for PF_6^- and $TFSI^-$based system, roughly corresponding the shift of the shoulder attached to the peak position in SAXS measurement (1.53-1.54 $Å^{-1}$). Similarly, another observation regards with increasing of Li-salt content AIMD results shows a good agreement with experimental intensities, where $\Delta S(q)$ for PF_6^- system is 0.09 and 0.11, and for $TFSI^-$ system 0.05 and 0.08 for AIMD and SAXS, respectively.

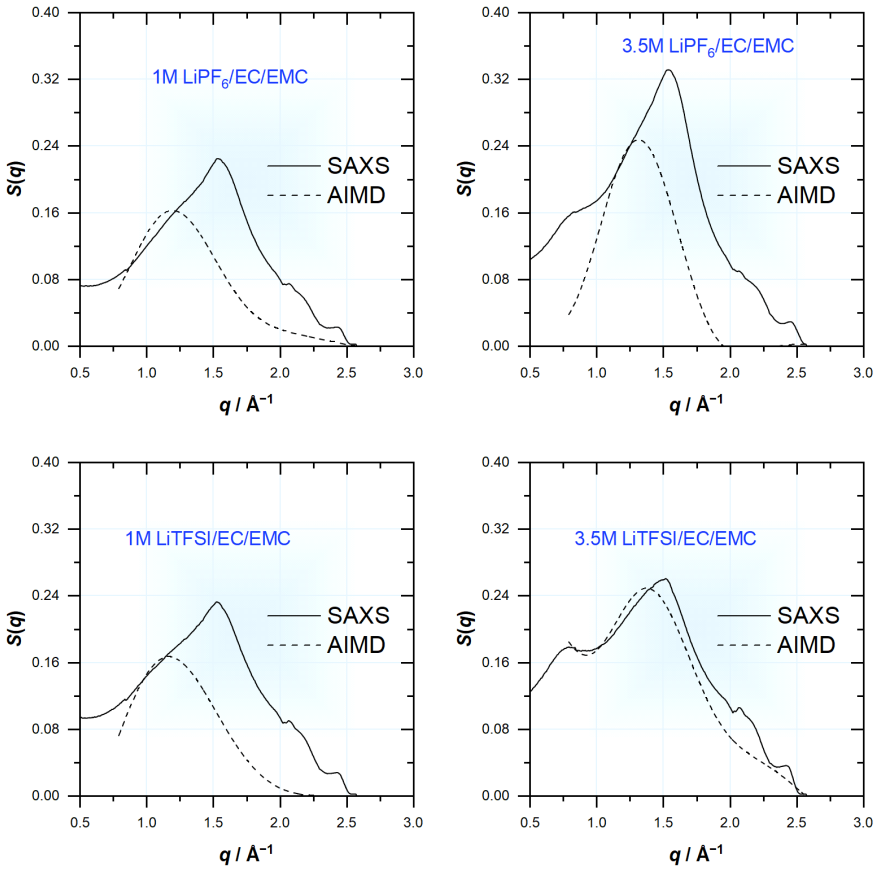


Figure 4. Comparison between the experimental SAXS and theoretical analysis of the structure factor, $S(q)$, for 1M and 3.5M $LiPF_6$/EC/EMC (*top*) and LiTFSI/EC/EMC (*bottom*) electrolytes, calculated by means of AIMD simulation at 298.15 K.

But the theoretical and experimental structure factors differ significantly, it may indicate that the model used to calculate the theoretical structure factor is incomplete or incorrect. Apart correct reproducibility of the intensity variation and peak positions against Li-salt concentration in electrolyte, the simulation method was not able to predict lower q values as we can see from the Figure 4. At this stage, one assumption can be made based on the relative low scale of the system considered that does not allow to predict lower q values. While AIMD simulations are a powerful tool for studying the properties of materials at the atomic level, they are limited by computational cost, time scale limitations when it comes to studying large systems.

2. Radial Distribution Function and Coordination Number

2.1 Lithium Solvation Environment

a) Interactions between Li ion and Anion.

The interaction between the Li-ion and the anion can have a significant impact on the solvation structure and dynamics. The anion can influence the Li solvation environment by influencing the electrostatic interactions between the Li ion and the solvent molecules.

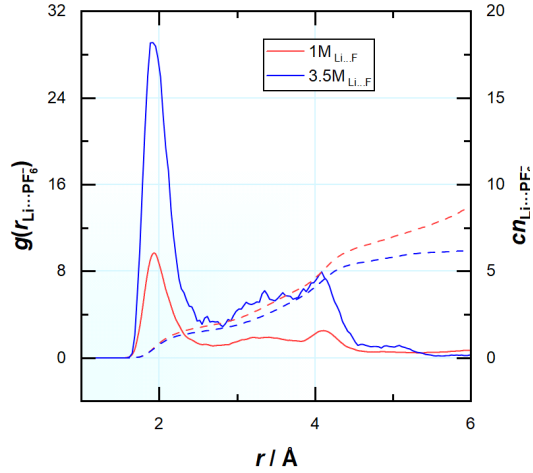


Figure 5a

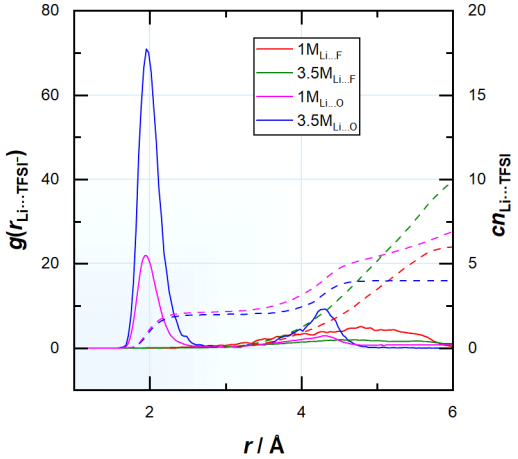


Figure 5b

Figure 5. Evolution of g(r) and coordination number graph for 1M and 3.5M of (a) $LiPF_6$ and (b) LiTFSI between Li and anion by means of AIMD simulation.

In LiTFSI system, the oxygen atom of the anion is a more favorable site for interaction with counterion compared to the fluorine atom. From both graphs in Figure 5 the peak position can be observed at similar distance around 1.91 Å and 1.92 Å for both the concentrations respectively. As the concentration of the system in increased in $LiPF_6$ its can be observed from Figure 5a that the intensity of peak position is increasing, similar to the results reported by Mallarapu[37]. The same tendency can also be observed in case of LiTFSI system. Increasing Li-salt content led to increasing the intensity of the peak position that might be reflected in higher probability of ionic aggregates formation. Also, Li^+ exhibit a higher coordination ability with respect to $TFSI^-$ than with PF_6^-. Increasing of the Li-salt content led to increasing the coordination ability of Li^+ to anion roughly by the same order.

b) Interaction between Li ion and Solvent

The interaction between the Li^+ and the solvent in LIB affects the Li solvation shell formation, the viscosity of the electrolyte, and the stability of the electrodes, which can have a significant impact on the performance and lifetime of the battery. In Figure 6, 7 is shown the RDF along with running coordination number for interaction between Li ion and EC/EMC solvents.

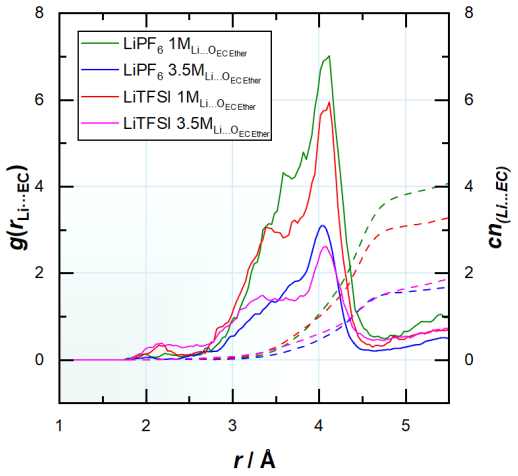


Figure 6a

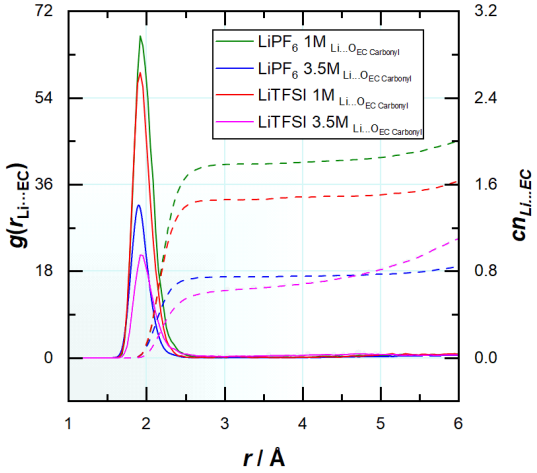


Figure 6b

Figure 6. Evolution of g(r) and coordination number graph for 1M and 3.5M of $LiPF_6$ and LiTFSI between Li and (a) Ether oxygen (b) Carbonyl oxygen of Ethylene Carbonate (EC) solvent by means of AIMD simulation

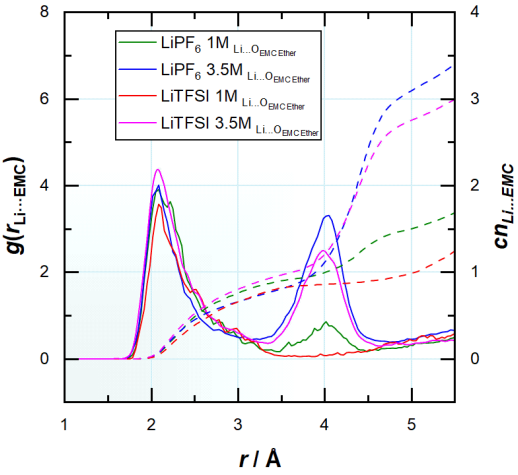


Figure 7a

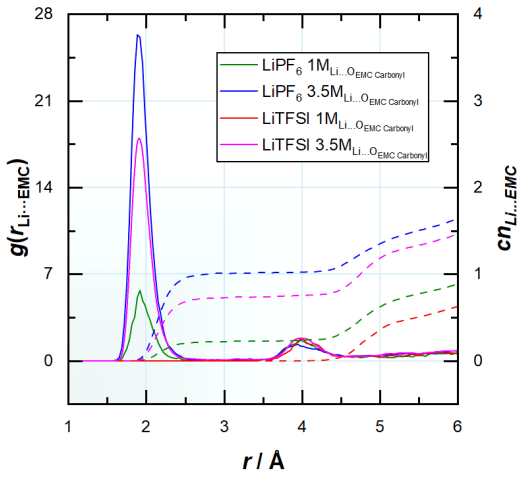


Figure 7b

Figure 7. Evolution of g(r) and coordination number graph for 1M and 3.5M of $LiPF_6$ and LiTFSI between Li and (a) Ether oxygen (b) Carbonyl oxygen of Ethyl methyl Carbonate (EMC) solvent by means of AIMD simulation

Among both the different type of oxygens in the solvent, carbonyl oxygen is more favorable, and both the solvents are localized around Li^+ by these sites. The same can be observed from the peak position, where the nearest peak position for ether oxygen –O– involved in ion-molecular interaction is 4.117 and 2.083 Å for EC and EMC, respectively, whereas for carbonyl oxygen O=C it found at *ca* 1.917 and 1.983 Å. The peak positions are almost coinciding for both 1M and 3.5M salt content, assuming that the nature of anion is weakly affected on the Li-solvent interactions. Another observation concerning the variation of salt content can be made based on solvent respective running coordination number behavior. Considering the most localized oxygen atom of O=C group, with the increasing of salt concentration the Li^+ coordination ability is decreasing in case of EC solvent that can be a sign of Li^+ desolvation, while for EMC the coordination ability following opposite trend. Such pattern corroborate well with previous observations by Mallarapu [37]. As for $LiPF_6^-$ based system, the coordination number decreased from 0.635 to 0.291 for EC solvent and for EMC, the coordination number varies from 0.085 to 0.358 A similar observation can be made for LiTFSI system. This reflects that increase in concentration leads to replacement of EC solvent by EMC solvent.

2.2 Hydrogen Bonding

a) Intermolecular Interactions

Intermolecular interactions, occurring between the solvent molecules, the lithium salt, and the electrode surfaces in the electrolyte, play a crucial role in the performance of lithium-ion batteries (LIBs) by influencing the transport of lithium ions and the stability of the electrodes. Because the interactions affect mobility and diffusivity of Li-ions and strong intermolecular interactions provide a protective layer on electrode surface which enhances stability and reduces degradation processes like solid electrolyte interphase formation. . One important type of intermolecular interaction is hydrogen bonding, which occurs between the solvent molecules and the lithium salt. The hydrogen bonding can affect the solvation shell formation around the Li^+, as well as the solubility and ionic conductivity of the electrolyte [53]

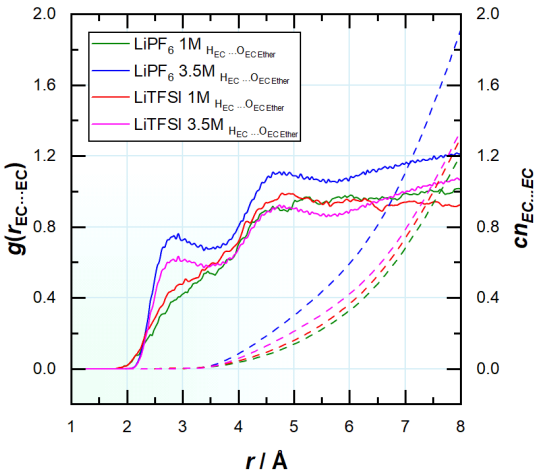


Figure 8a

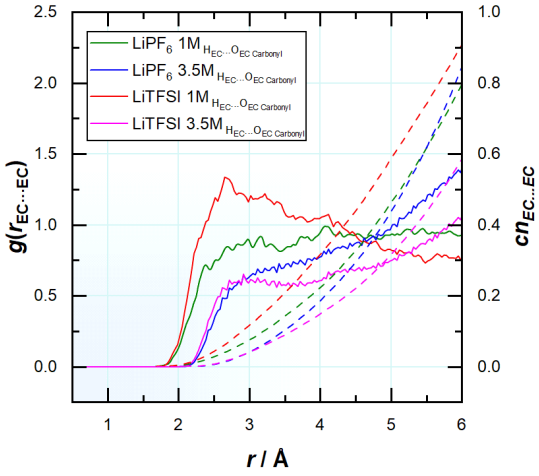


Figure 8b

Figure 8. Evolution of g(r) and coordination number graph of intermolecular interactions between H and (a) Ether oxygen (b) Carbonyl oxygen among Ethylene Carbonate (EC) solvents for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of AIMD simulation

We have considered all the possible intermolecular hydrogen bonding interactions between EC and EMC. In order to understand the criteria of hydrogen bonding formation we the geometrical criteria of hydrogen bonding formation proposed by Steiner [54] . According to these criteria, if the distance between hydrogen atom and the electron accepting atom of interacting molecule is >2.2 Å, such hydrogen bonds considered as a weakly bonded interaction. In Figure 8, 9 the RDF and running coordination number is plotted for EC···EC and EC···EMC interactions. Tendency of weak hydrogen bonding formation observed between EC molecules, and between EC and EMC due to higher accepting ability of O=C group, compared to −O− group. For same concentration of salt, LiTFSI system show stronger hydrogen bonding interaction compared to $LiPF_6$. Greater hydrogen bonding interaction among the solvents reflect in high viscosity of electrolyte, which leads to lower mobility of anions and, hence, effects on the ionic conductivity. Another observation is increasing Li-salt content leading to the weakening for hydrogen bonding interactions as a result of strengthening of interionic interactions, particularly in PF_6^- based system.

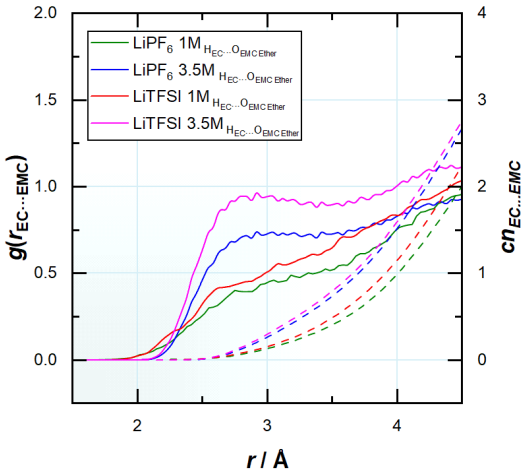


Figure 9a

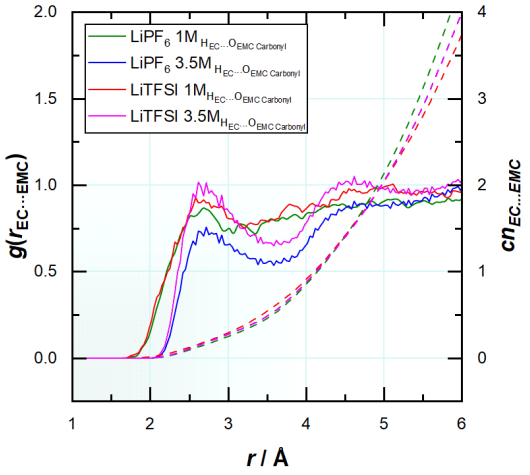


Figure 9b

Figure 9. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O (a) Ether oxygen (b) Carbonyl oxygen of Ethylene Carbonate (EC) solvent and Ethyl Methyl Carbonate (EMC) for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of AIMD simulation

In Figure 10, 11 the RDF and running coordination number is illustrated for interactions between EMC···EMC and EMC···EC solvents. Considering the similar criteria for hydrogen bonding interactions it is also observed to be a tendency of weak hydrogen bonding formation between these solvents. Among the methyl hydrogen ($-CH_3$) and methylene ($-CH_2$) moieties, hydrogen atoms of $-CH_2$ group shows a tendency of stronger interaction[55]. From Figure 11 it can be observed that interactions are slightly stronger for LiTFSI than $LiPF_6$ especially for EMC···EC interactions. In this particular case, the observed minor changes in the interaction suggest that the presence of anions has a minimal effect on the hydrogen bonding interactions. These interactions are weaker compared to other solvent interactions, e.g., EC···EC. Intermolecular interactions between solvent molecules are crucial criteria which effect the viscosity and electrode-electrode stability. Balancing the strength of the hydrogen bonding interactions and other intermolecular interactions within the electrolyte is important to achieve optimal battery performance.

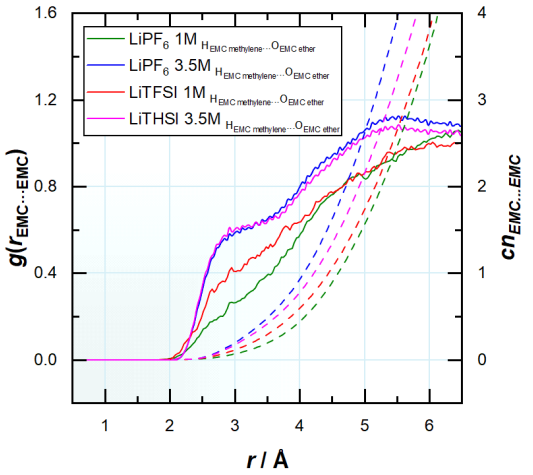


Figure 10a

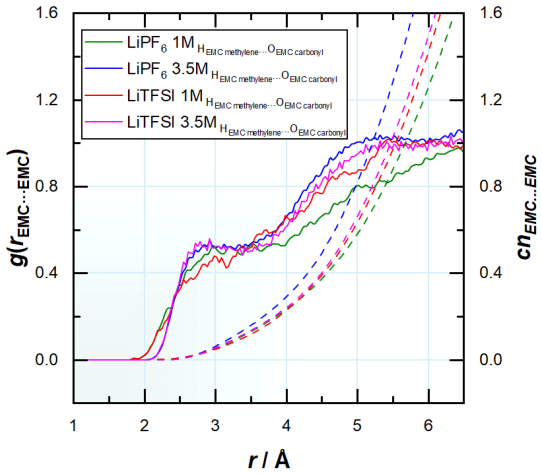


Figure 10b

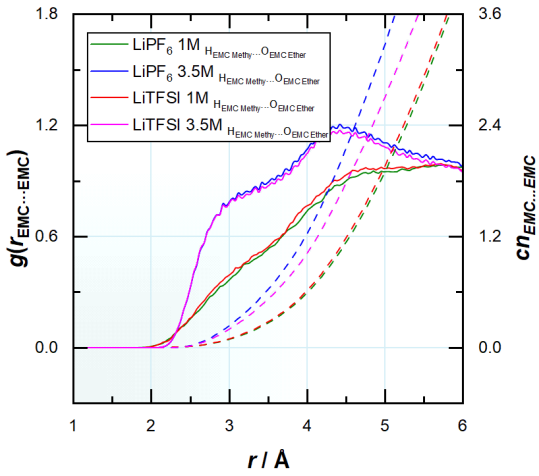


Figure 10c

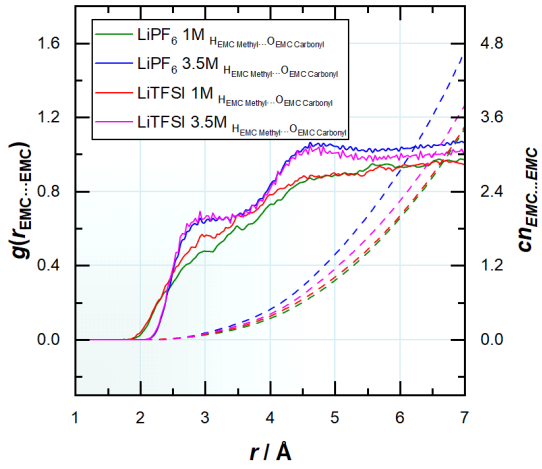


Figure 10d

Figure 10. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O among Ethyl Methyl Carbonate (EMC) of (a) methylene-ether (b) methylene-carbonyl (c) methyl-ether (d) methyl-carbonyl for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of AIMD simulation

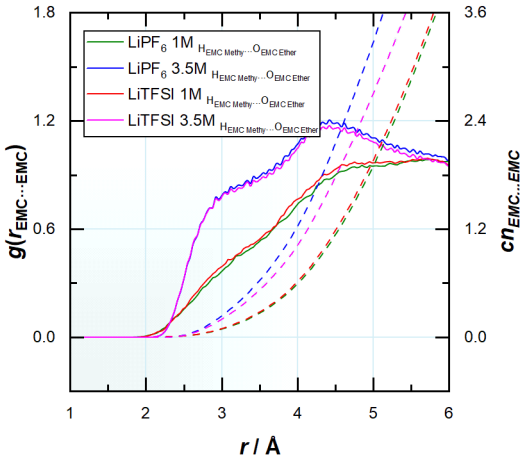


Figure 11a

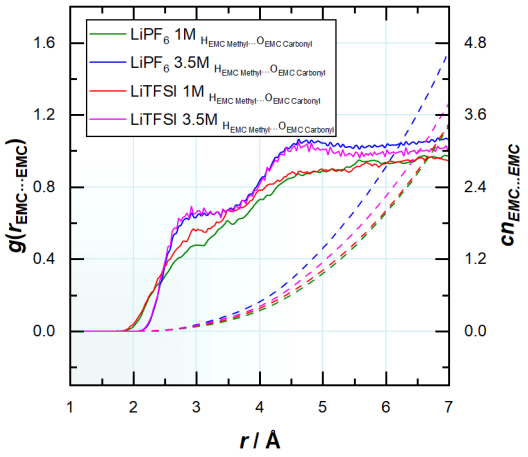


Figure 11b

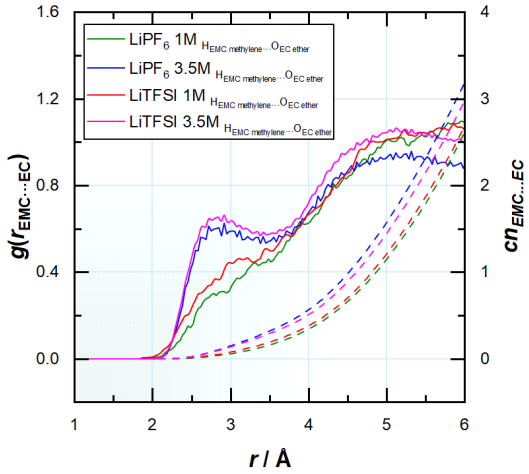


Figure 11c

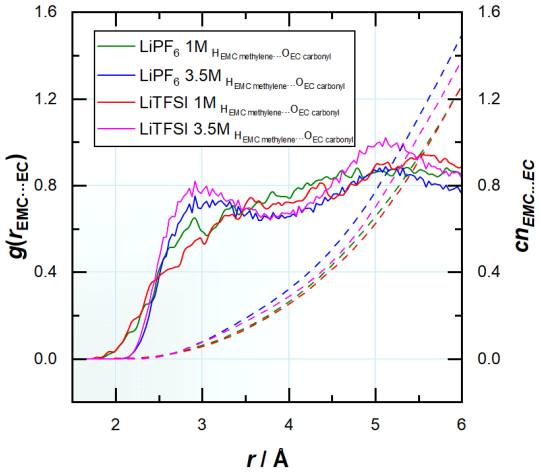


Figure 11d

Figure 11. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O of Ethyl Methyl Carbonate (EMC) and Ethylene Carbonate (EC) solvent of (a) methyl-ether (b) methyl-carbonyl (c) methylene-ether (d) methylene-carbonyl for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of AIMD simulation

b) Ion Molecular Interactions

The transportation of Li ions and the stability of the electrodes in the electrolyte are influenced by the ion-molecular interactions involving solvent molecules, lithium salt, and electrode surfaces. Specifically, the interaction between solvent molecules and the lithium salt plays a significant role in hydrogen bonding. This hydrogen bonding affects the formation of the solvation shell around lithium ions and also impacts the solubility and ionic conductivity of the electrolyte.

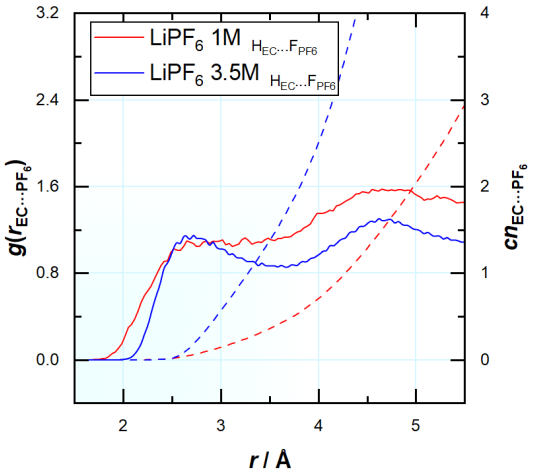


Figure 12a

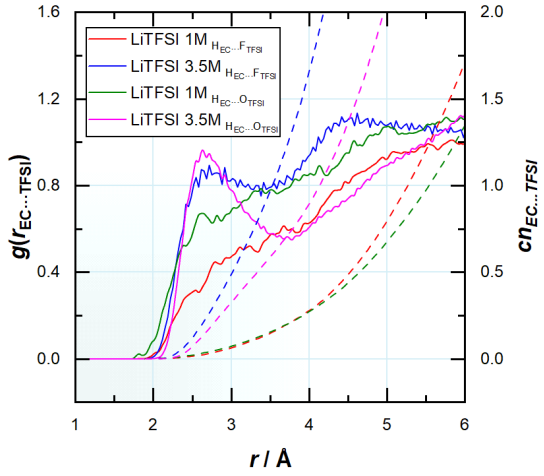


Figure 12b

Figure 12. Evolution of g(r) and coordination number graph of ion interactions between Ethylene Carbonate (EC) solvent and anion of 1M and 3.5M of (a) $LiPF_6$ and (b) LiTFSI by means of AIMD simulation

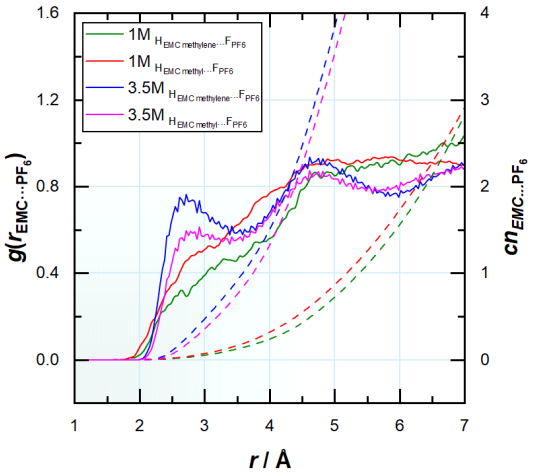


Figure 13a

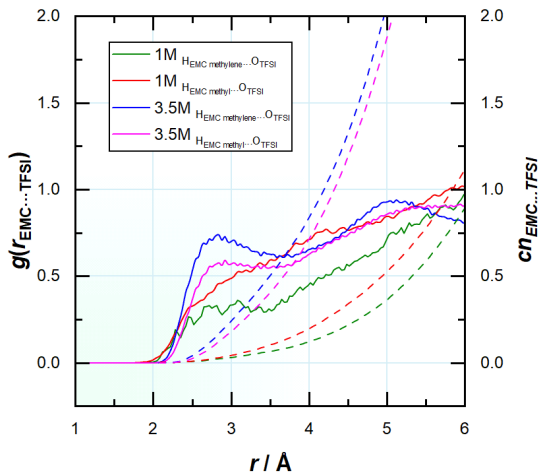


Figure 13b

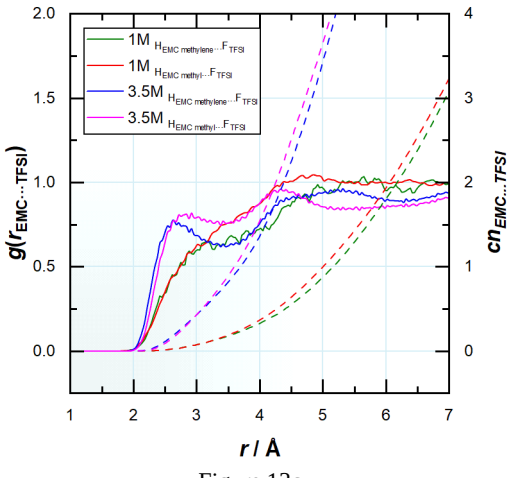


Figure 13c

Figure 13. Evolution of g(r) and coordination number graph of ion interactions between Ethyl methyl Carbonate (EMC) solvent and anion of 1M and 3.5M of (a) $LiPF_6$ and of (b), (c) LiTFSI by means of AIMD simulation

We have analyzed the RDF and running coordination number from both the salt systems between solvent and anion interactions as it shown in Figure 12, 13. Tendency of weak ion-molecular hydrogen bonding formation observed for both systems, that particularly pronounced for PF_6^-. In case of LiTFSI, as the salt concentration is increasing the ability to form these interactions is increasing with EC solvent. Comparing both the solvents, EC interacts more strongly with the anions than EMC for both the electrolyte system. Also, with increasing of Li-salt content the strength of ion-molecular hydrogen bonds is also increasing in opposite the intermolecular hydrogen bonding. As a result of strengthening of interionic interactions it results in desolvation of the Li^+. For EMC anions are observed to interact stronger with methyl group at 1M and methylene group at 3.5M of Li-salt content

Even though we were able to discuss the Li ion solvation environment and hydrogen bonding interactions, the inability to produce lower q values, thereby, not able to match the peak of SAXS experimental analysis structure factor is observed. The main reason being the limitation in system size using the AIMD analysis. Classical MD can be used to simulate an essentially larger systems, typically consisting of thousands or even millions of atoms, as it only requires the use of a potential energy function. We considered to do the similar simulation using classical MD with a larger structural system as mentioned in Table 1 and sis the same analysis as we did in AIMD.

B. Classical MD Approach

1. Structure Factor analysis

As it was mentioned, structure factor is a key parameter in comparing theoretical and experimental analyses of the material structure organization. In Figure 14 we have compared the structure factor of all the four systems and EC/EMC solvents against the SAXS experimental data. It can be seen that there is match in the peak position and intensity of peak for all the systems. With increasing of Li-salt content, classical MD results shows a reasonably high agreement with experimental intensities, where $\Delta S(q)$ for PF_6^- system is 0.11, and for $TFSI^-$ system 0.04 in both classical MD and SAXS analysis. The observed deviation of curves might be associated with ionic aggregate formation in the experimental system, or a

much larger system is needed to compensate the deviation. Regardless, unlike AIMD, classical MD well corroborate with the SAXS experimental measurements.

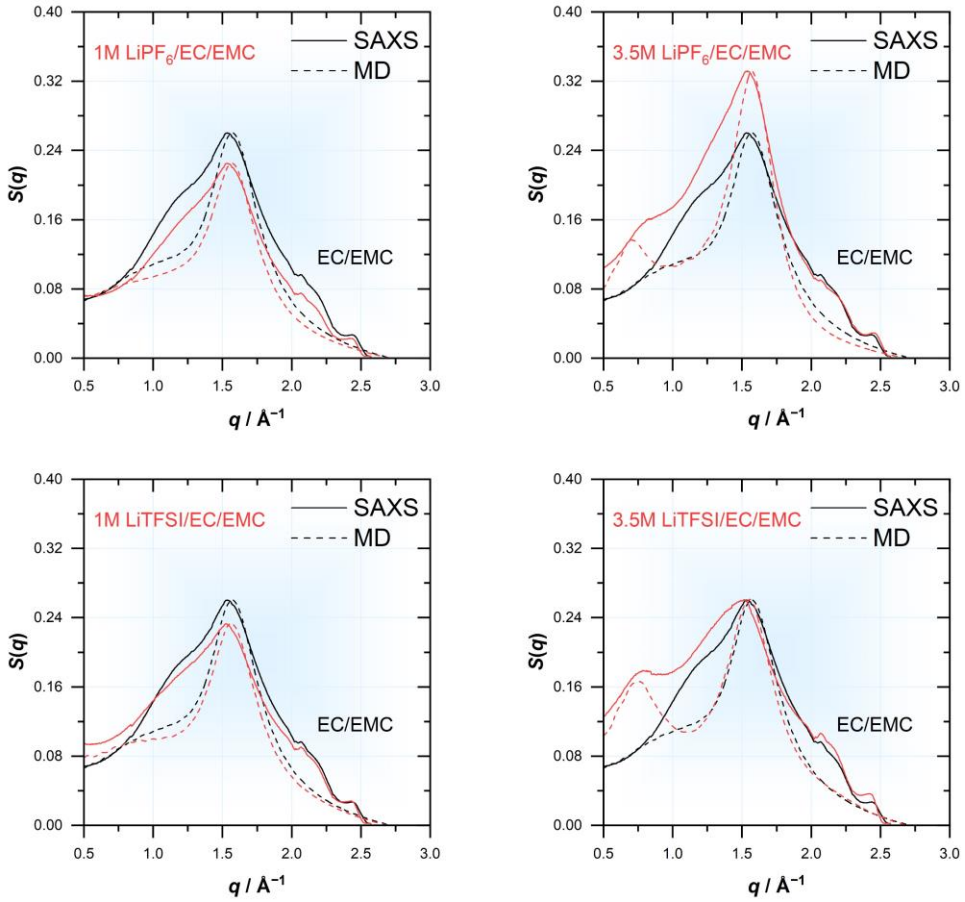


Figure 14. Comparison between the experimental SAXS and theoretical analysis of the structure factor, $S(q)$, for 1M and 3.5M $LiPF_6$/EC/EMC (*top*) and LiTFSI/EC/EMC (*bottom*) electrolytes, as well as neat EC/EMC binary mixture used as a reference, calculated by means of classical MD simulation at 298.15 K.

2. Radial Distribution Function and Coordination Number

2.1 Lithium Solvation Environment

a) Interactions between Li ion and Anion.

RDF and running coordination number are plotted between Li^+ ion and anion interactions. The anion can influence the solvation environment between the Li ion and the solvent molecules, as well as the solvent molecules orientation and mobility. From Figure 15 its observed that in LiTFSI system oxygen is most favorable site for interaction compared to fluorine which was a similar observation in AIMD.

Increasing Li-salt content enhances the strength of interionic interactions and Li^+ coordination, particularly with PF_6^-. This can be observed from the change in coordination number from 0.007 to 0.248 and 0.007 to 0.107 for both $LiPF_6$ and LiTFSI systems, respectively. It was a similar observation made by Mallarapu [37] in their work on $LiPF_6$-based system. However, the difference that is observed for both the simulation methods for cation anion interaction is the intensity of the peak. The intensity of the peaks produced by AIMD based systems are much greater compared to the classic MD system in Figure 15, which means that compared to AIMD, classical MD clearly reflects a stronger tendency of Li^+ to solvation rather than aggregation.

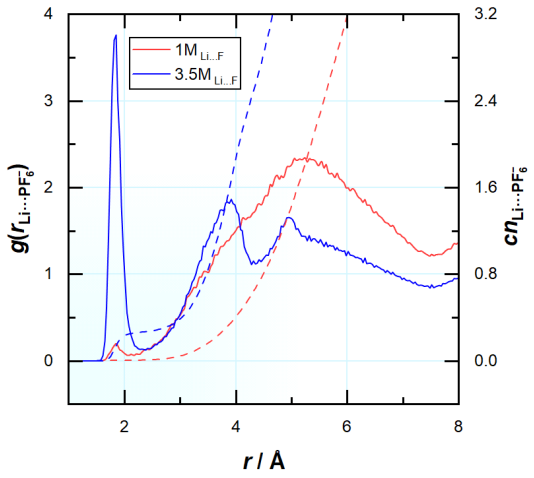


Figure 15a

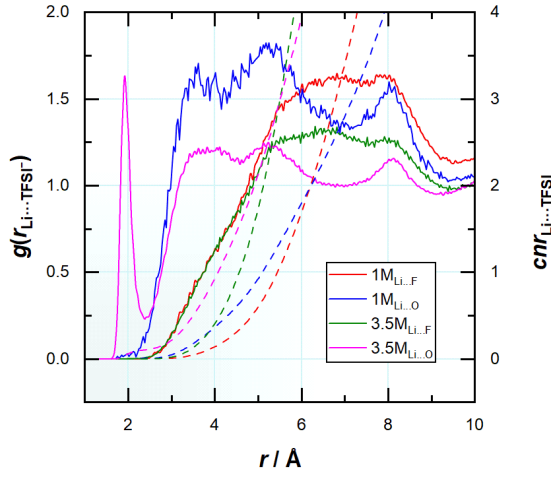


Figure 15b

Figure 15. Evolution of g(r) and coordination number graph for 1M and 3.5M of (a) $LiPF_6$ and (b) LiTFSI between Li and anion by means of MD simulation

b) Interaction between Li ion and Solvent

The interaction of Li^+ and solvent in a LIB influences solvation shell formation, electrolyte viscosity, and electrode stability which have impact on the performance of a battery. We considered Li-ion interaction with carbonyl and ether oxygens of the solvent solution and plotted RDF and running coordination number as it shown in Figure 16, 17. The solvent molecules are localized around Li^+ ion by oxygen of O=C. Seeing the peak position for all the systems, regardless of the change in anion the peak positions are same for both ether oxygen interaction (3.8 Å) and carbonyl oxygen (1.8 Å). In this case anions don't have much influence on the Li solvent interactions. Comparing the Li coordination ability to

both solvents, with the increasing of salt concentration the coordination number of EC for O=C goes from 0.635 to 0.291 and 0.618 to 0.197 for $LiPF_6$ and LiTFSI systems, respectively, whereas, for EMC it goes from 0.085 to 0.358 and 0.176 to 0.316 for $LiPF_6$ and LiTFSI systems. Thus, unlike EMC, increasing of Li-salt content lead to decreasing of Li^+ coordination ability to EC and, hence, desolvation of the cation and EC is getting replaced by EMC .The same is confirmed by Mallarapu [37] and his team in his work. Compared to AIMD, classical MD results clearly is more sensitive to the variation in electrolyte composition.

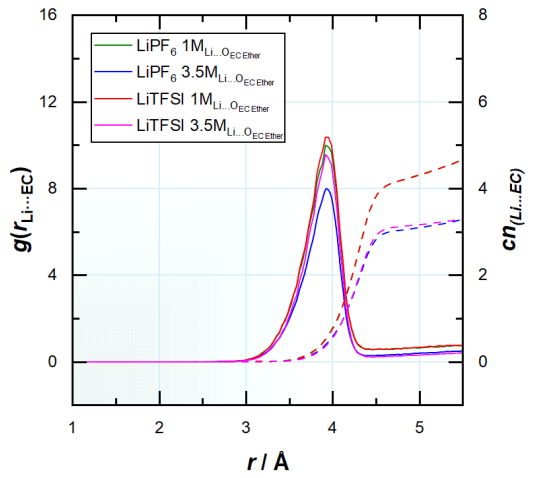


Figure 16a

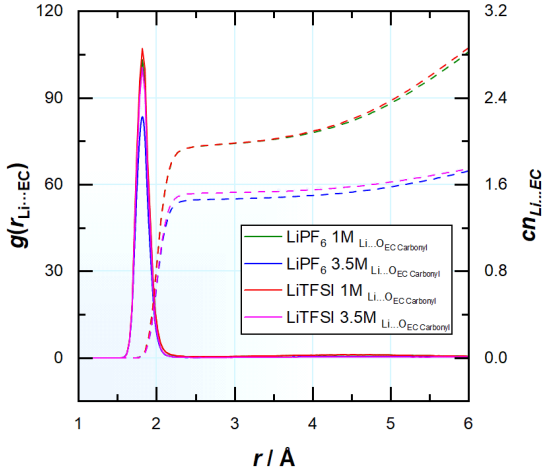


Figure 16b

Figure 16. Evolution of g(r) and coordination number graph for 1M and 3.5M of $LiPF_6$ and LiTFSI between Li and (a) Ether oxygen (b) Carbonyl oxygen of Ethylene Carbonate (EC) solvent by means of MD simulation

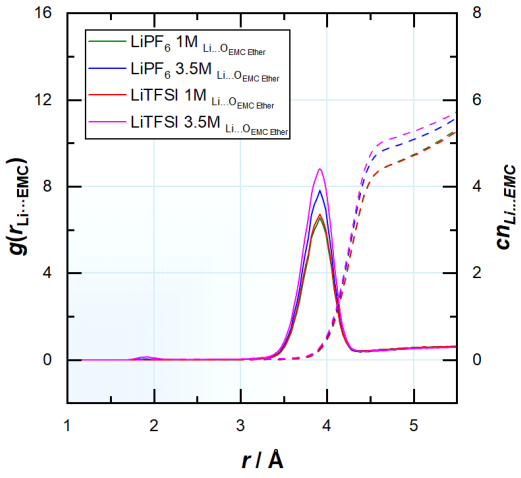


Figure 17a

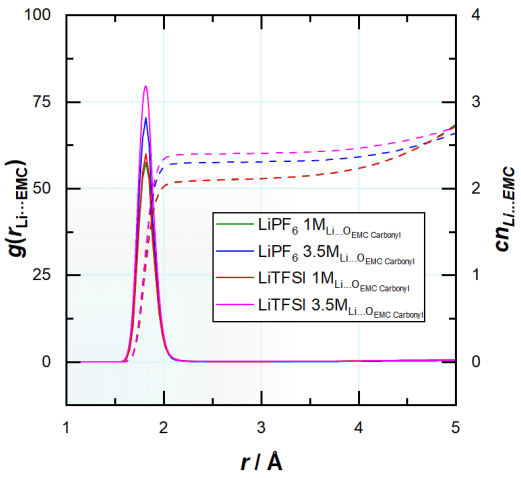


Figure 17b

Figure 17. Evolution of g(r) and coordination number graph for 1M and 3.5M of $LiPF_6$ and LiTFSI between Li and (a) Ether oxygen (b) Carbonyl oxygen of Ethyl methyl Carbonate (EMC) solvent by means of MD simulation

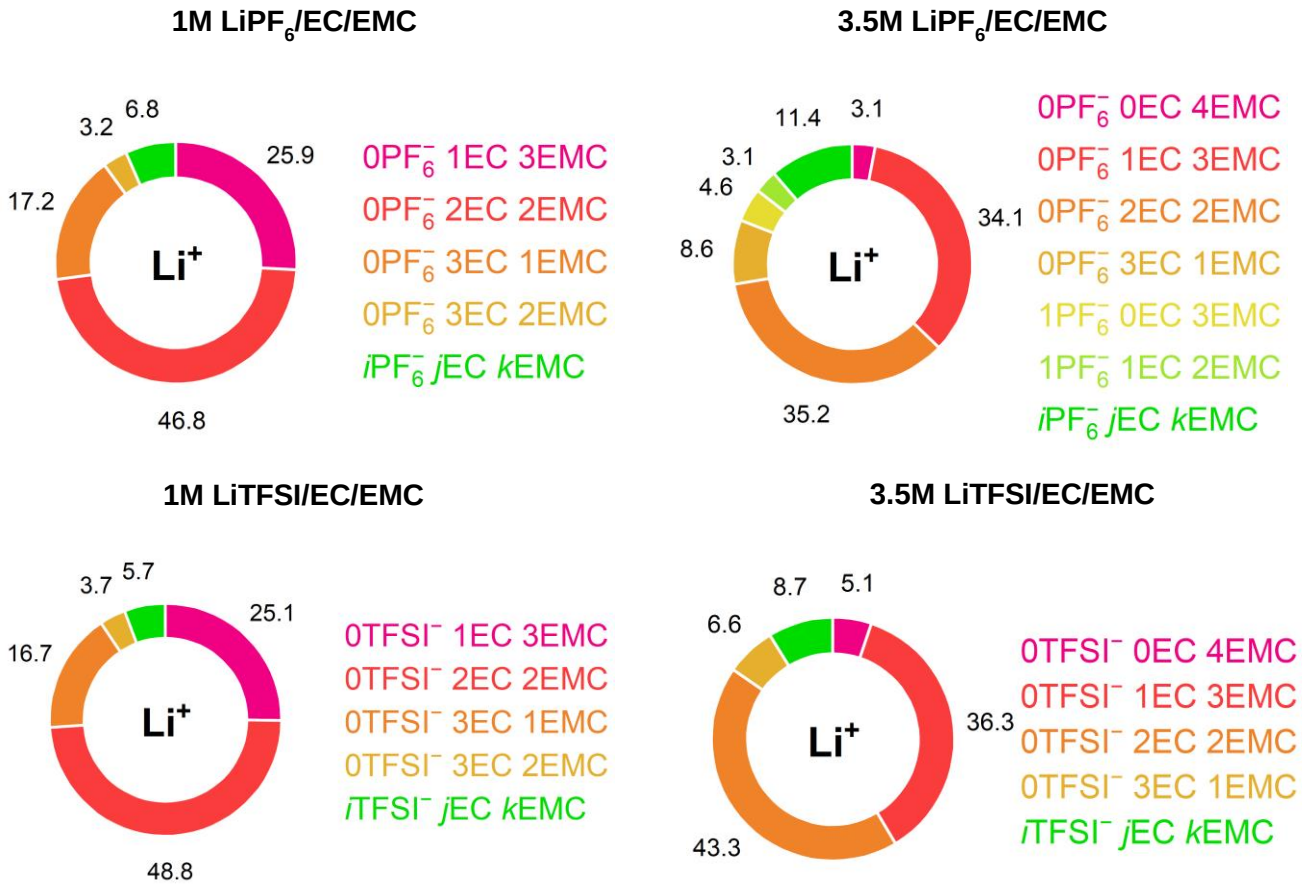


Fig. 18. Probability distribution (in %) of ion-molecular complexes in the first solvation shell of Li in $LiPF_6$/EC/EMC (Top) and LiTFSI/EC/EMC (Bottom) electrolytes

Here in figure 18, there is a pictorial representation of probability distribution of ion molecular complexes in first solvation shell. Increasing Li-salt content lead to strengthening of interionic interactions, while the percentage of PF_6^- (2.01-4.57%) and $TFSI^-$ (1.15-1.47%) in Li first solvation shell is essentially lower compared to EC/EMC molecules. At 1M of Li-salt the Li^+ is entirely solvated by EC and EMC. Increasing Li-salt lead to redistribution of the solvation structure with a dominant contribution of EMC.

2.2 Hydrogen Bonding

a) Intermolecular Interactions

Intermolecular interactions as mentioned previously are vital for the performance of LIBs, as they occur between the solvent molecules, the Li salt, and the electrode surfaces in the electrolyte, influencing the transport of lithium ions and the stability of the electrodes. Hydrogen bonding is a significant type of

intermolecular interaction that takes place between the solvent molecules and the lithium salt in the electrolyte. These hydrogen bonding interactions have a considerable impact on the formation of the solvation shell around the lithium ions, as well as on the solubility and ionic conductivity of the electrolyte.

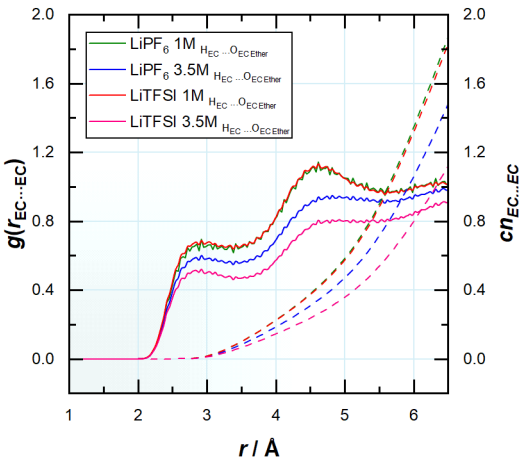


Figure 19a

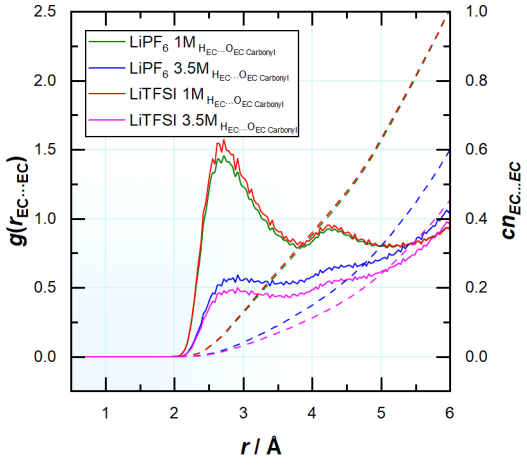


Figure 19b

Figure 19. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O among Ethylene Carbonate (EC) solvents of (a) Ether oxygen and (b) Carbonyl oxygen for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of MD simulation

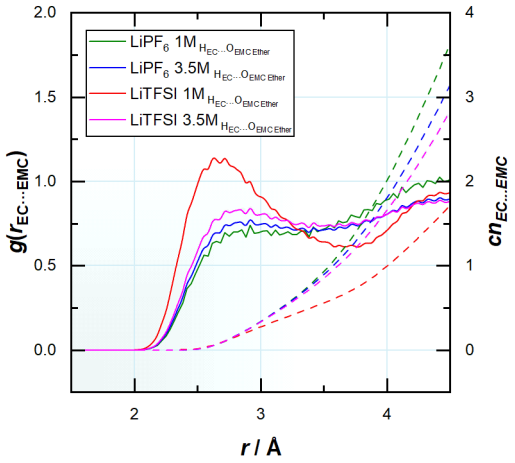


Figure 20a

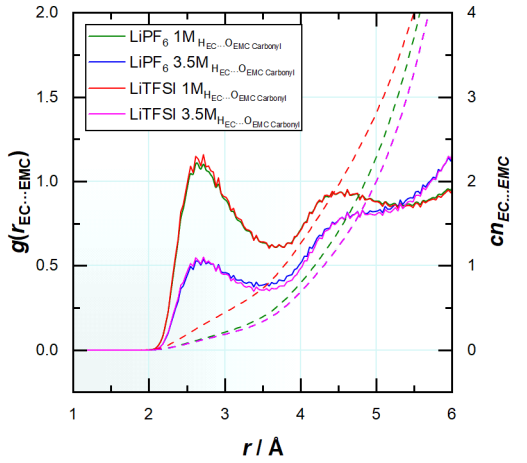


Figure 20b

Figure 20. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O of Ethylene Carbonate (EC) solvent and Ethyl Methyl Carbonate (EMC) of (a) Ether oxygen and (b) Carbonyl oxygen for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of MD simulation

Figure 19 and Figure 20 shows evolution of RDF between EC···EC and EC···EMC interactions with respect to the system composition and nature of the constituents. RDF plot focus on the interaction between hydrogens and oxygens of the solvents. As observed oxygen of the carbonyl group is tend have stronger interaction. The first peak position RDF shown is approximately 2.9 Å which shows a tendency of weak

hydrogen bonding interaction according to Steiner[54]. One common observation that can be made in all the plots are the electrolytes with lower concentration 1M of salt tend to have stronger hydrogen bonding compared to the 3.5M concentration. Increasing the strength of interionic interactions diminish intermolecular hydrogen bonding, particularly in case, where EC molecules stands a donor of electrons. At lower concentrations hydrogen bonding interactions are not much affected by the nature of anion. But a slightly better interactions for $LiPF_6$ can be seen at higher concentration system.

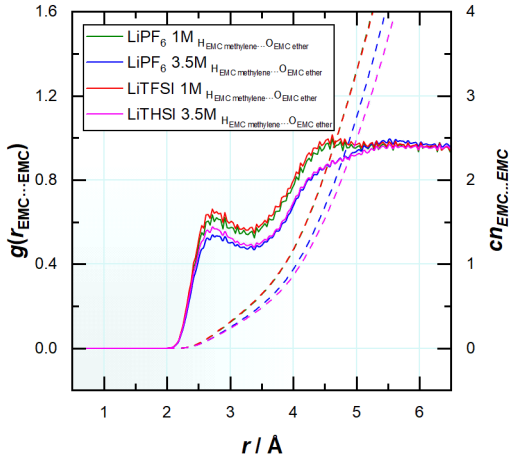


Figure 21a

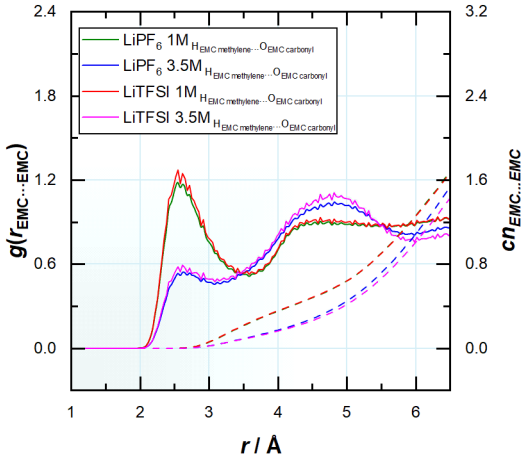


Figure 21b

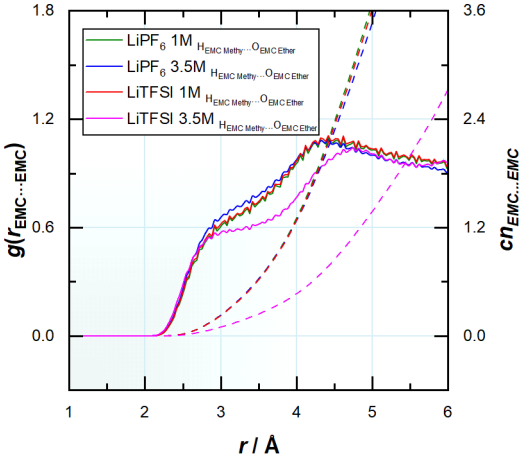


Figure 21c

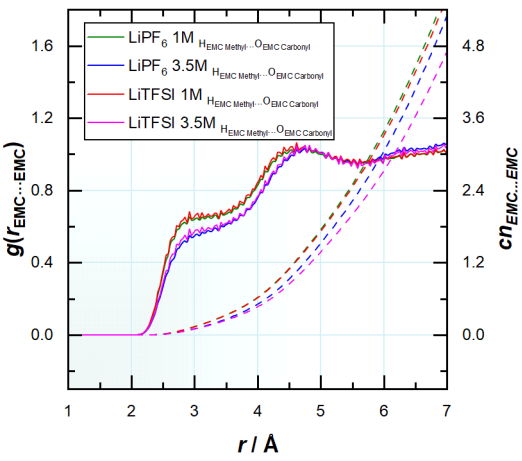


Figure 21d

Figure 21. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O among Ethyl Methyl Carbonate (EMC) (a) methylene-ether (b) methylene-carbonyl (c) methyl-ether (d) methyl-carbonyl for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of MD simulation

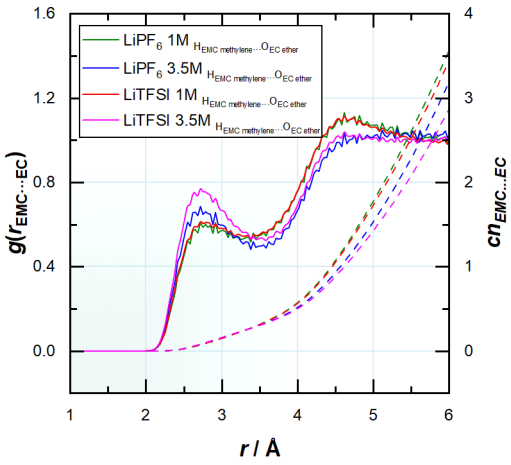


Figure 22a

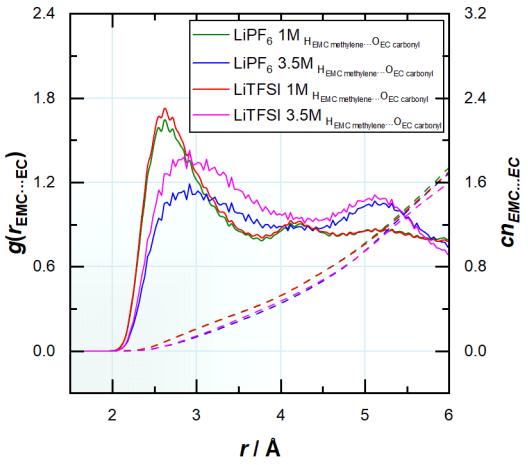


Figure 22b

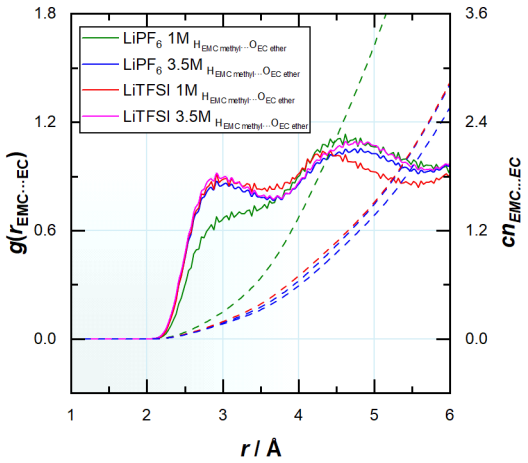


Figure 22c

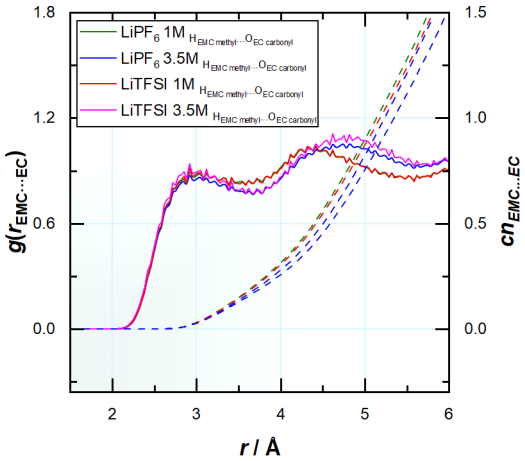


Figure 22d

Figure 22. Evolution of g(r) and coordination number graph of intermolecular interactions between H and O of Ethyl Methyl Carbonate (EMC) and Ethylene Carbonate (EC) solvent (a) methylene-ether (b) methylene-carbonyl (c) methyl-ether (d) methyl-carbonyl for 1M and 3.5M of $LiPF_6$ and LiTFSI by means of MD simulation

The EMC solvent contains hydrogen of both the methylene and methyl group. In Figure 21, 22 we considered its interaction with oxygens of both solvents (EMC···EMC and EMC···EC). Weaker tendency of hydrogen bonding formation is observed. Interaction between methylene hydrogen and carbonyl oxygen are stronger which is observed for both EMC···EMC and EC···EC. In addition to that the methyl group hydrogen are less electronegative [63]. Moreover, in Figure 21 we can observe that the nature of anions is ineffective in the hydrogen bonding interactions between EMC···EMC. The capability for hydrogen bonding formation is much pronounced in EC than in EMC, due to the higher electron donating/accepting ability. Increasing the strength of interionic interactions diminish intermolecular hydrogen bonding,

particularly in case, where EC molecules stands a donor of electrons. Apart the same predictive ability of both computational techniques, MD simulation more sensitive to the variation of electrolyte composition.

b) Ion Molecular Interactions

The functioning of LIBs is significantly affected by molecular interactions that take place between the solvent molecules, the lithium salt, and the electrode surfaces in the electrolyte. These interactions play a crucial role in the transport of lithium ions and the stability of the electrodes. Hydrogen bonding is a particularly important type of intermolecular interaction that occurs between the solvent molecules and the lithium salt. This hydrogen bonding can affect the formation of the solvation shell around the lithium ions, and affect the solubility and ionic conductivity of the electrolyte.

Figure 23a

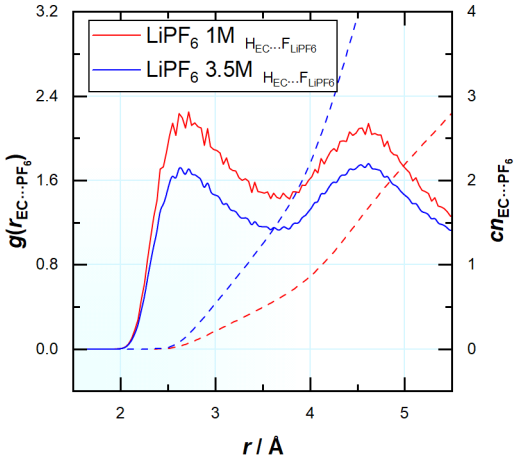


Figure 23b

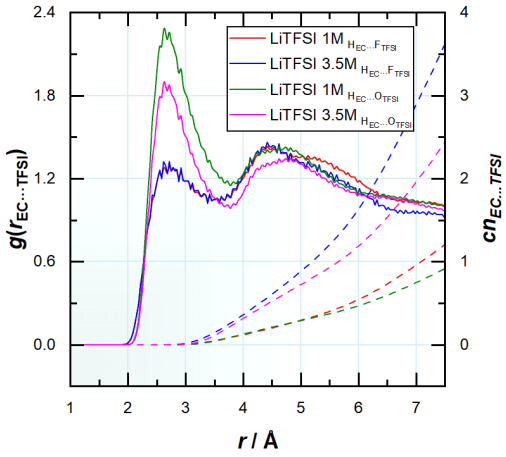


Figure 23. Evolution of g(r) and coordination number graph of ion interactions between Ethylene Carbonate (EC) solvent and anion of 1M and 3.5M of (a) $LiPF_6$ and (b) LiTFSI by means of MD simulation

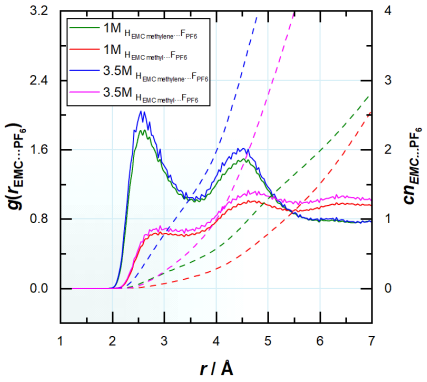


Figure 24a

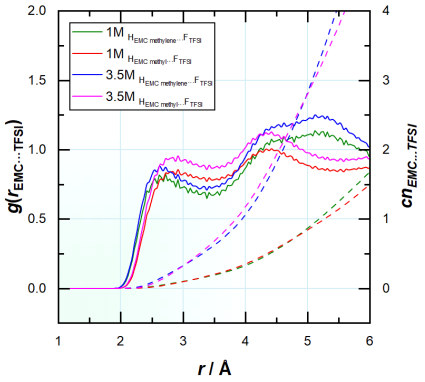


Figure 24b

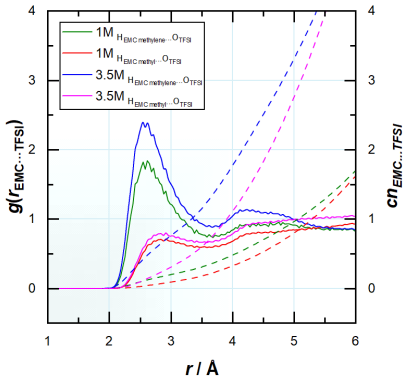


Figure 24c

Figure 24. Evolution of g(r) and coordination number graph of ion interactions between Ethyl methyl Carbonate (EMC) solvent and anion of 1M and 3.5M of (a) $LiPF_6$ and (b), (c) LiTFSI by means of MD simulation

To understand the ion molecular interactions the RDF and running coordination number are plotted as it presented in Figure 23, 24. Similar to previous hydrogen bonding interactions, here as well a weak interaction tendency is observed, that particularly pronounced for PF_6^-. In LiTFSI the solvent interaction with O atom is dominating over the F atom and hydrogen from methyl group show stronger interactions for EMC solvent. As said earlier in the intermolecular interaction, here as the salt concentration is increased Li-salt content disrupt intermolecular interactions in a favor of interionic and ion-molecular ones, particularly for EMC···anion.

VI. Conclusions

In present work an effect of the salt concentration, as well as the structure of anion was investigated in electrolytes based on the 1 and 3.5M LiPF6 and LiTFSI in the binary mixtures of EC/EMC with constant volume ration of 0.44. For this purpose, a multiscale approach using AIMD and classical MD techniques was applied to describe a local structure organization on the atomistic level. To validate the obtained results, first, the structure factor was analyzed against experimental SAXS measurements. It was found that classical MD analysis produce a reasonably high convergence with experiment then the AIMD approach. Such difference, particularly for lower *q* values, was explained by the limitation in the system size, on one hand, while the mismatch in peak position and intensities by the limited number of counterions (AIMD) in the system, and different extent of ionic association that strongly depend on the potential model parameters (MD), on the other.

Next, Li^+ solvation environment and hydrogen bonding formation in terms of interionic, intermolecular and ion-molecular interactions, demonstrated and compared against the two techniques, anions, and different salt compositions. It was found that the systems computed by AIMD featured by higher probability of ionic association (aggregate formation), whereas the same systems simulated with classical MD showed a stronger tendency of Li^+ solvation rather than aggregation. Despite the fact that AIMD generally provides much accurate results, the classical MD was found to be more sensitive to the variation of electrolyte composition, which is in line with experimental observations.

Both salt systems were then compared to each other, and the effects of salt content variation was studied. Therefore, a stronger interaction between Li^+ ions and anions, especially the PF_6^- anion, was observed. However, the coordination ability of LiTFSI was found to be dominant when considering anions. Increasing Li-salt content lead to the strengthening of interionic interaction, and the nature of the anion weakly affected Li-solvent interactions. As Li-salt content increased, EC was replaced by EMC in Li^+ solvation shell. Overall, these findings provide valuable insights into the behavior of LiTFSI and $LiPF_6$-based electrolytes with binary (EC/EMC) solvent mixtures, highlighting the strengths and limitations of AIMD and classical MD simulation methods in analyzing these systems.

The solvation study and hydrogen bonding interaction study conducted in this work provided valuable insights into the ionic conductivity and coulombic efficiency, both of which significantly impact the performance of batteries. The quality of the interaction between Li ions and the solvent can affect the mobility of Li ions in the electrolyte solution, and thus the coulombic efficiency and ionic conductivity of the LIB. It was observed that at lower concentrations, solvent molecules strongly bind to Li ions for LiTFSI, resulting in reduced mobility of Li ions and decreased coulombic efficiency. Conversely, solvent molecules weakly interact with Li ions for $LiPF_6$, resulting in higher mobility of Li ions and better coulombic efficiency [14].

Similarly, hydrogen bonding can affect the solvation shell formation around the lithium ions, as well as the solubility and ionic conductivity of the electrolyte. For example, solvents like EC have a higher tendency for hydrogen bonding formation with Li^+, which decreases the mobility of ions and increases the viscosity of the electrolyte. However, it was observed that as the salt concentration increased, the EC solvent interactions weakened, leading to improved Li^+ solvation and mobility, resulting in better ionic conductivity.

Overall, the nature of the electrolytes clearly influences the electrochemical performance of batteries, which can be explained in terms of the organization and interaction of the species in solution and linked to the physicochemical properties of solvents. These findings emphasize the importance of understanding the solvation behavior and hydrogen bonding interactions in electrolyte solutions to improve the performance of batteries.

VII. References

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